

Musaliar College of Engineering & Technology

MUSALIAR COLLEGE P.O., PATHANAMTHITTA - 689 653



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LABORATORY RECORD

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Sri/Smt.....of.....Semester

Class of (Reg No.....)

of.....Branch

in the.....Laboratory

during the academic year 20 - 20

Name of Examination:

Internal Examiner

External Examiner

Staff in-charge

DEPARTMENT OF COMPUTER APPLICATIONS

VISION

“To produce competent and dynamic professionals in the field of Computer Applications to thrive and cater the changing needs of the society through research and education”.

MISSION

To impart high quality technical education and knowledge in Computer Applications. To introduce moral, ethical and social values to Computer Application students. To establish industry institute interaction to enhance the skills of Computer Application students. To promote research aimed towards betterment of society.

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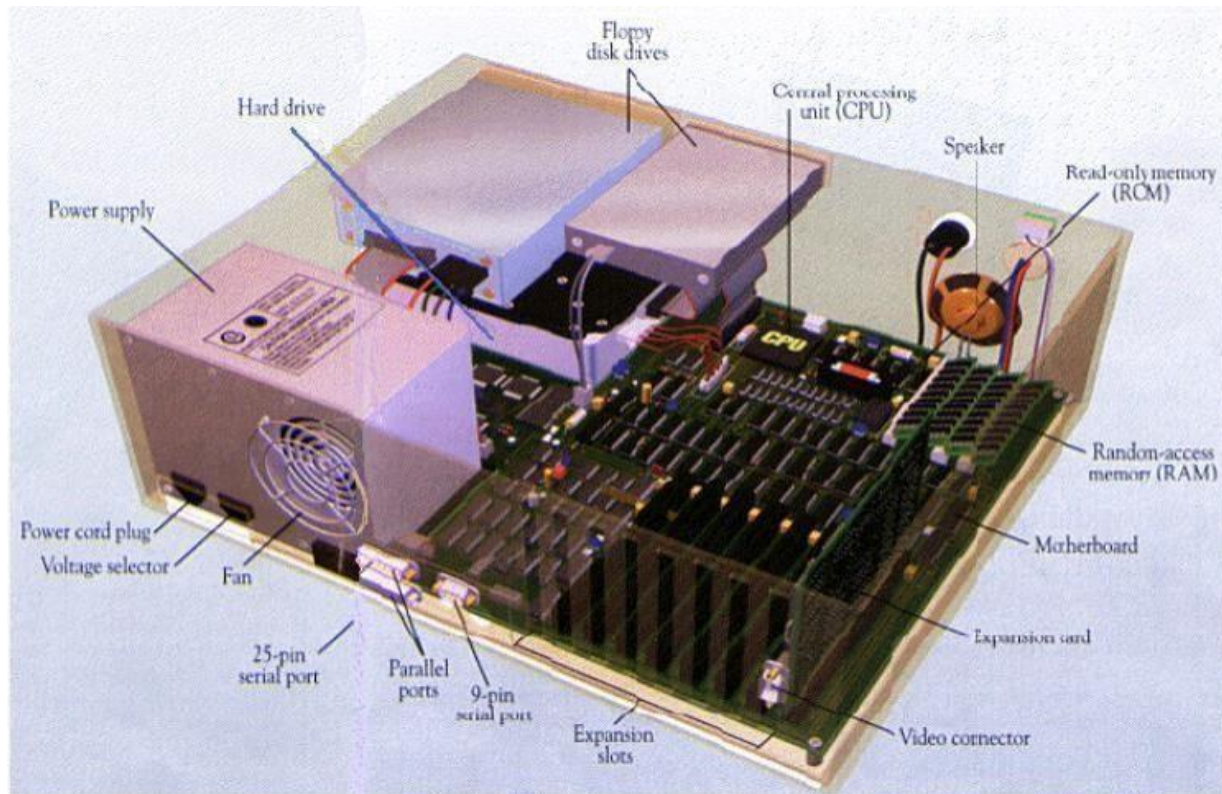
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EXPERIMENT 1

INTRODUCTION TO COMPUTER HARDWARE

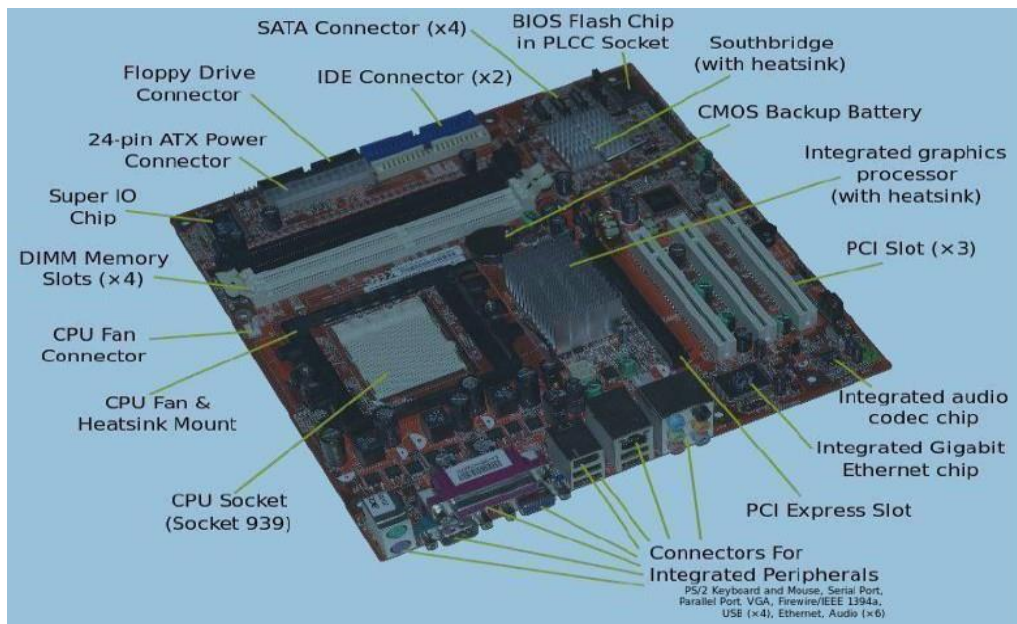
AIM

Physical identification of major components of a computer system such as mother board, RAM modules, daughter cards, bus slots, SMPS, internal storage devices, interfacing ports.



2. Motherboard

The motherboard has been an integral part of most personal computers. The motherboard is a common circuit board. Each circuit card performs a unique function in the computer and gets its power from the socket. The motherboard has been an integral part of most personal computers. The motherboard is a common circuit.



Types of connectors used in motherboard

1. IDE connector
2. SATA connector
3. Floppy connector
4. Power supply connector
5. Keyboard connector [PS/2 or USB]
6. Mouse connector [PS/2 or USB]
7. VGA connector [Monitor data connector]
8. Parallel Port
9. Serial Port
10. Audio connector
11. LED connector
12. FAN connector
13. USB port

2. CPU

The CPU, or processor, is the heart of your computer no matter what type (PC, Server, and Laptop). There are many brands for processors such as Intel and Athlon all with different processors for your computer.



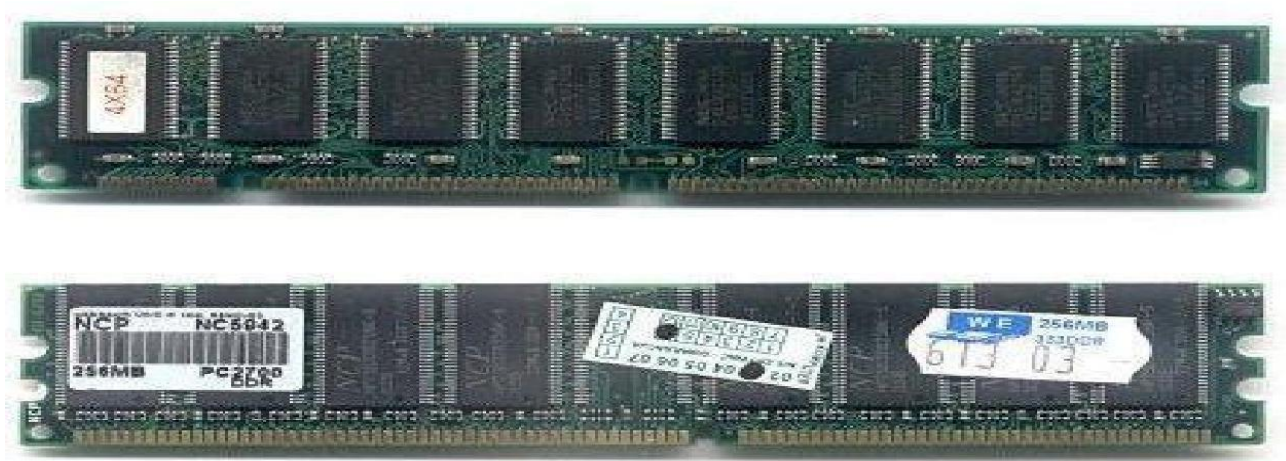
3. RAM MODULES:

In computing, a **memory module** or **RAM (random-access memory) stick** is a printed circuit board on which memory integrated circuits are mounted. Memory modules permit easy installation and replacement in electronic systems, especially computers such as personal computers, workstations, and servers. The first memory modules were proprietary designs that were specific to a model of computer from a specific manufacturer. Later, memory modules were standardized by organizations such as JEDEC and could be used in any system designed to use them.

Types of memory module include:

- TransFlash Memory Module
- SIMM, a single in-line memory module

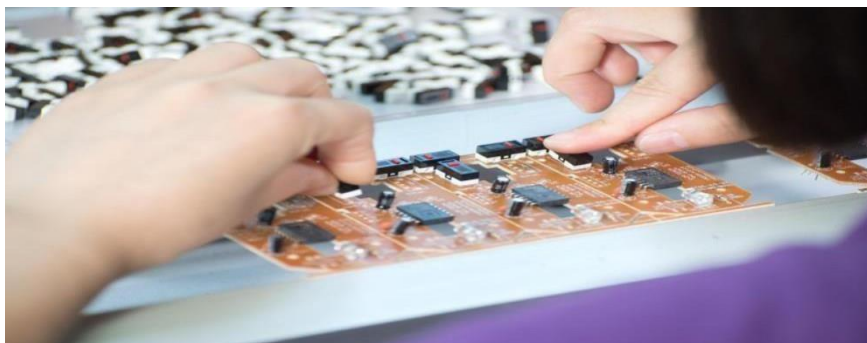
- DIMM, dual in-line memory module
- Rambus memory modules are a subset of DIMMs, but are normally referred to as RIMMs
- SO-DIMM, small outline DIMM, a smaller version of the DIMM, used in laptops



Two types of DIMMs (dual in-line memory modules): a 16 pin SDRAM module (top) and a 184- pin DDR SDRAM module (bottom)

4. DAUGHTER CARDS

A daughter card or daughterboard is a type of circuit board that gets added to an existing one. Its name is appropriate for its use, since it is connected to a “motherboard” or “main board.” The motherboard is the primary circuit board for a device. It is usually in the device as it is shipped from the factory. A daughter card may be added later.



5. Hard-Disk Drive

A hard drive stores all your files and information in a permanent form unlike storing it in RAM (which is temporary). The larger your hard disk (drive) the more information about files you can store.



Types of hard drive

- SATA Hard Drives
- External Hard Drives and Solid State Drives (SSD)
- USB Flash Drives
- Gaming Hard Drives
- PS4 Hard Drives

6. CD-ROM:

The CD-Rom is reads CD's. CD-Rom completely stands for Compact Disk Read Only Memory. CD's have much more data than a floppy disk. Using CD-RW you can make your own CD's and use them more like a floppy disk.

CD-Rom:



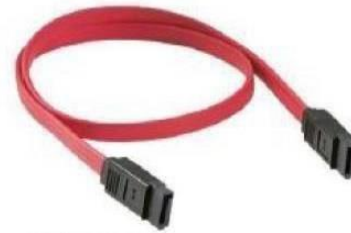
CD-disk



IDE cable :



SATA cable



7. SMPS

SMPS or switched mode power supply is an electronics device that is used to convert AC to DC, AC to AC, DC to DC and DC to AC voltages. The circuit consists of transformer, rectifier, voltage regulator and a filter. The input supply is turned On and Off by MOSFETs and outputs a high efficiency, regulated voltage. The SMPS used in computers is of AC to DC type supply.



Power-IN. The power-IN connector as shown in the figure is the input for MAINS supply. A power cable is inserted here, the other end of which is connected to mains supply. The input supply gets converted to DC supply.

Power-OUT. The power-OUT connector is connected directly to the Power-IN connector from inside the supply unit. It supplies the same AC supply that is fed to power-IN socket. The power OUT connector is used to give supply to monitors or any display unit.

FAN. If you look at the back side of Computer- SMPS, you will find a FAN at the right side. The FAN as you can see, blows the air out and is only used to dissipate the internal heat from the SMPS since the switching is done at high frequencies which create a lot of heat inside.

ATX connector. As you can see in the image, it is a 24-pin female connector which is used to supply DC supply to the motherboards. Various color-coded wires connect to this connector and each colored wire supplies distinct DC voltage



ATX-12V connector. Latest SMPS power supplies are accompanied by an extra 4-pin connector which supplies 12 volts to energize the central processing unit and other components of the motherboard.



AT Connectors. Earlier motherboards used to support AT connectors(6-pin each) also called P8 and P9 connectors to supply power to these motherboards(upto 486 boards).



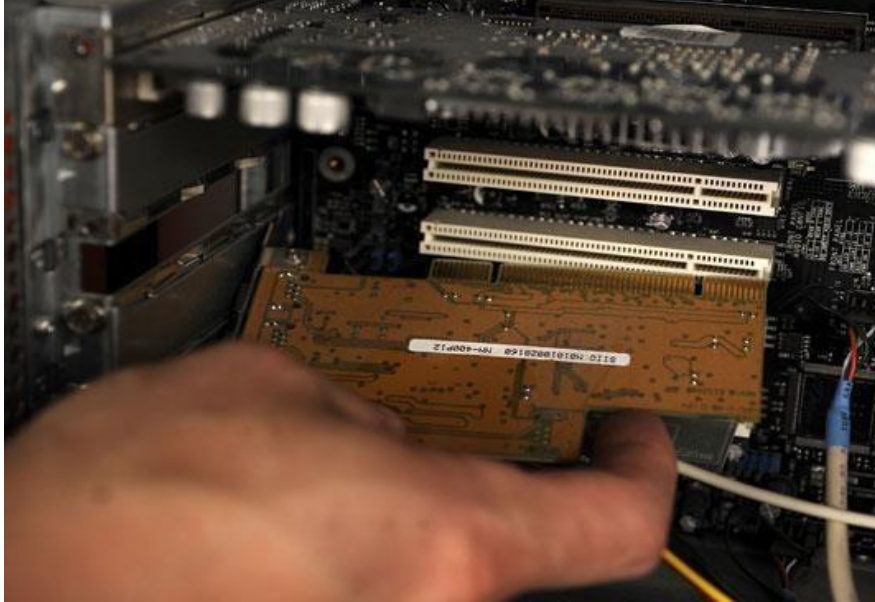
PIN connectors. There are multiple 4-pin connectors that draw out from the SPMS unit. These connectors are used to supply DC power to various peripherals of computer like a floppy disk drive, hard disk drive or DVD-writers.



SATA-output connector. To feed the power to latest SATA hard drives, these connectors are used.

8. INTERNAL/EXPANSION BUSES

We can expand your PC internally by adding additional circuitry boards. Those boards, or expansion cards, plug directly into expansion slots on the motherboard (as shown in this figure). So, we can expand your computer system by adding options not included with the basic PC. Here are the types of expansion slots your PC may have:



- **PCI Express:** The best type of expansion slot to have in your PC is the PCI Express, also written as PCIe. Without boring you, the PCI Express type of expansion slot communicates with the motherboard, and therefore with the microprocessor, both quickly and efficiently.
- **PCI:** The PCI slot is the most common form of internal expansion for a PC.
- **AGP:** This type of expansion slot was specifically designed to deal with graphics adapters. In fact, AGP stands for Accelerated Graphics Port. Older PCs may sport this expansion slot, but the best video cards use PCI Express.
- **ISA:** The most ancient type of expansion slot is the ISA, which stands for (get this) Industry Standard Architecture. That's because it never really had a name until another, better type of expansion slot came along. ISA slots hang around to be compatible with older expansion cards.

9. EXTERNAL PORTS

1. **VGA:** The **Video Graphics Array (VGA) connector** is standard connector used for computer video output. Originating with the 1987 IBM PS/2 and its VGA graphics system, the 15-pin connector went on to become ubiquitous on PCs, as well as many monitors, projectors and high definition television sets



High-Definition Multimedia Interface (HDMI):It is a proprietary audio/video interface for transmitting uncompressed video data and compressed or uncompressed digital audio data from an HDMI-compliant source device, such as a display controller, to a compatible computer monitor, video projector, digital television, or digital audio device. HDMI is a digital replacement for analog video standards.



3. **USB:** It is a standard cable connection interface for personal computers and consumer electronics devices. USB stands for Universal Serial Bus, an industry standard for short-distance digital data communications. USB ports allow USB devices to be connected to each other with and transfer digital data over USB cables. They can also supply electric power across the cable to devices that need it.



10. MONITORS

A monitor is an electronic output device that is also known as a **video display terminal (VDT)** or a **video display unit (VDU)**. It is used to display images, text, video, and graphics information generated by a connected computer via a computer's video card.

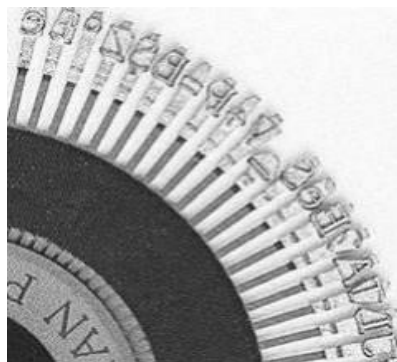
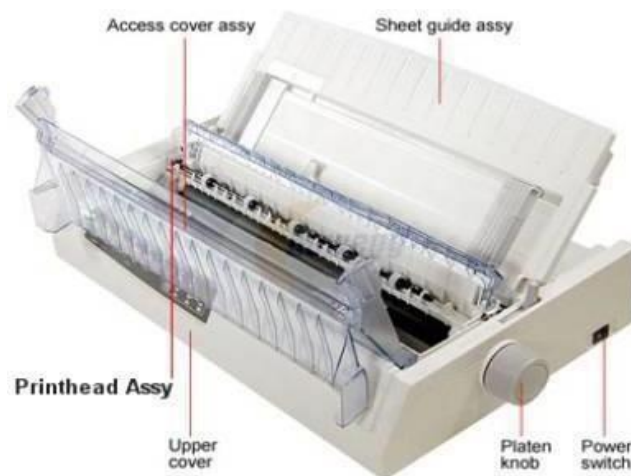
Types of Monitors

1. Cathode Ray Tube (CRT) Monitors
2. Flat Panel Monitors(Liquid Crystal Display & Gas Plasma Display)
3. Touch Screen Monitors

4. LED Monitors
5. OLED Monitors
6. DLP Monitors
7. TFT Monitors
8. Plasma Screen Monitors

10. PRINTERS

Printers are commonly used output devices that produce a hard copy of document stored in electronic form, i.e they put information from computer on to paper. There are various kinds of printers available today like Impact printers, Bubble-jet printers, Laser printers, Thermal printers etc.



11. SCANNERS

Scanner is a device that scans images, printed text, and handwriting etc and converts it to digital form or image. It is so named because the data is converted one line at a time or scanned down the page as the scanning head moves down the page.

Types of scanners

1. Flatbed Scanners



2. Sheetfed Scanners



3. Handheld Scanners



Result

Physical identification of major components of a computer is done successfully

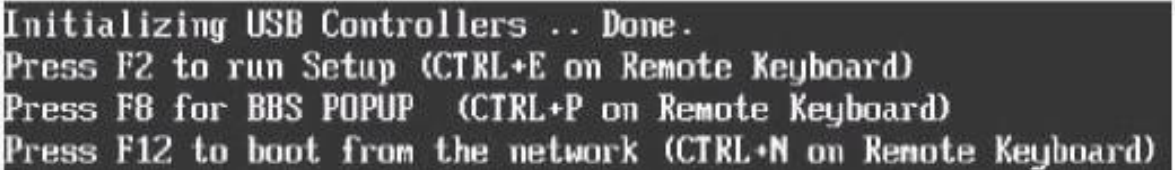
b. AIM

Familiarization of OS Installations

The BIOS Setup utility reports system information and can be used to configure the server BIOS settings. BIOS have a Setup utility stored in the BIOS flash memory. The configured data is stored in the system's battery- backed CMOS RAM.

To access BIOS configuration screens and to change the system's parameters complete the following steps:

1. Power on or power cycle the server.
2. To enter the BIOS Setup utility, press the F2 key while the system is performing the power-on self-test (POST)
3. Press F2 to Run Setup Prompt

4. 

5. When BIOS is started, the main BIOS Setup utility top-level screen appears. This screen provides seven menu options across the top of the screen.
6. Use the left and right arrow keys to select the different menu options
7. To select an option on a top-level screen, use the up and down arrow keys to scroll up and down the options presented.
8. Only options that can be modified are highlighted when you press the up and down arrow keys
9. If a field can be modified, as you select the option, user instructions for modifying the option appear in the right column of the screen
10. If a field is a link to a sub-screen, instructions to press the Enter key to access the sub screen appear in the right column.
11. Modify the setup field and press the Esc key to save the changes and exit the screen. Some screens present a confirmation dialog box that enables unwanted changes to be retracted.
12. On sub-screens that only provide configuration information and cannot be modified, press the Esc key to exit the screen.
13. To continue modifying other setup parameters, repeat Step 3 through Step 6. Otherwise, go to Step 8
14. Press and release the right arrow key until the Exit menu screen appears.
15. Follow the instructions on the Exit menu screen to save or discard your changes and exit

the BIOS Setup utility.

16. BIOS Setup Screens Overview: Screen Description.

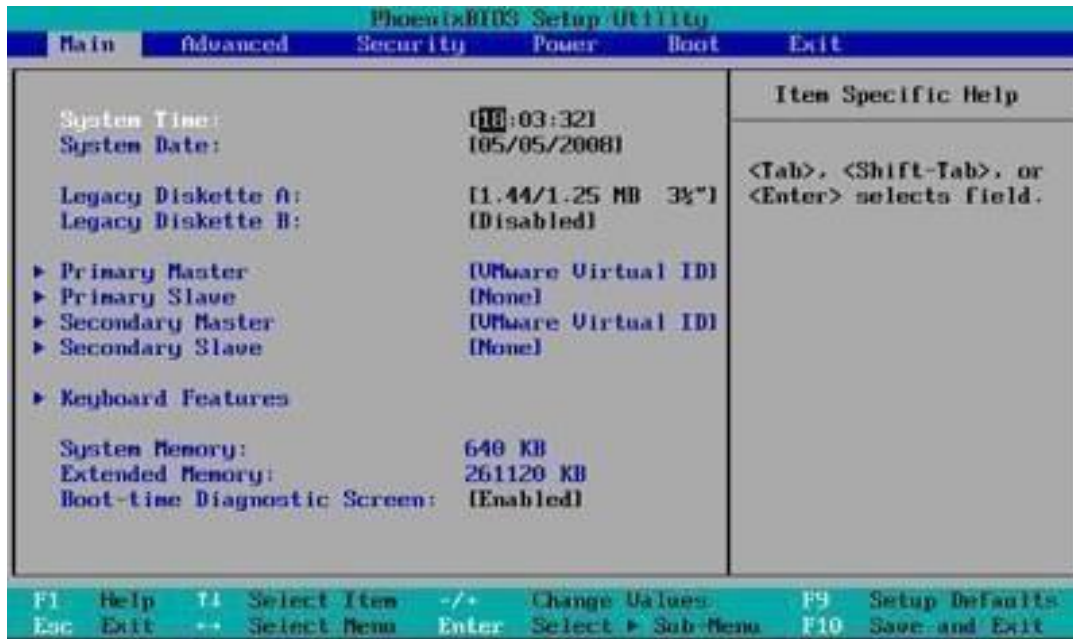
BIOS Setup Screens Overview:

Screen	Description
Main	General product information, including BIOS type, processor, memory, and time/date.
Advanced	Configuration information for the CPU, memory, IDE, Super IO, trusted computing, USB, PCI, MPS and other information.
PCI	Configure the server to clear NVRAM during system boot.
Boot	Configure the boot device priority (storage drives and the DVD-ROM drive).
Security	Set or change the user and supervisor passwords.
Chipset	View the configuration of server chipsets.
Exit	Save changes and exit, discard changes and exit, discard changes, or load optimal or fail-safe defaults.

BIOS Setup Utility Menu Screens :

The BIOS Main screens provide general product information, including BIOS, processor, system memory, and system time/date.

BIOS Setup Utility: Main



Advanced Setting:



Exit - Save Configuration Changes and Exit Confirmation:



BIOS Power-On Self-Test (POST) Events:

At system startup, the BIOS perform a power-on self-test that checks the hardware on your server to ensure that all components are present and functioning properly.

INSTALL AND CONFIGURE : OS INSTALLATION

Ensure that your computer meets or exceeds the minimum system requirements to run Xubuntu Linux:

Minimum requirements :

- gigahertz (GHz) or faster 32-bit (x86) or 64-bit (x64) processor*
- 1 gigabyte (GB) RAM (32-bit) or 2 GB RAM (64-bit)
- 16 GB available hard disk space (32-bit) or 20 GB (64-bit)
- DirectX 9 graphics device with WDDM 1.0 or higher driver
- Super VGA 800x600 Display Adapter
- CD or DVD-ROM
- Keyboard and mouse, or other pointing devices

- Network Interface Adapter required for Internet and Network Connectivity

Steps to install Xubuntu

Step 1: Create a live Xubuntu USB or disk if it is not available

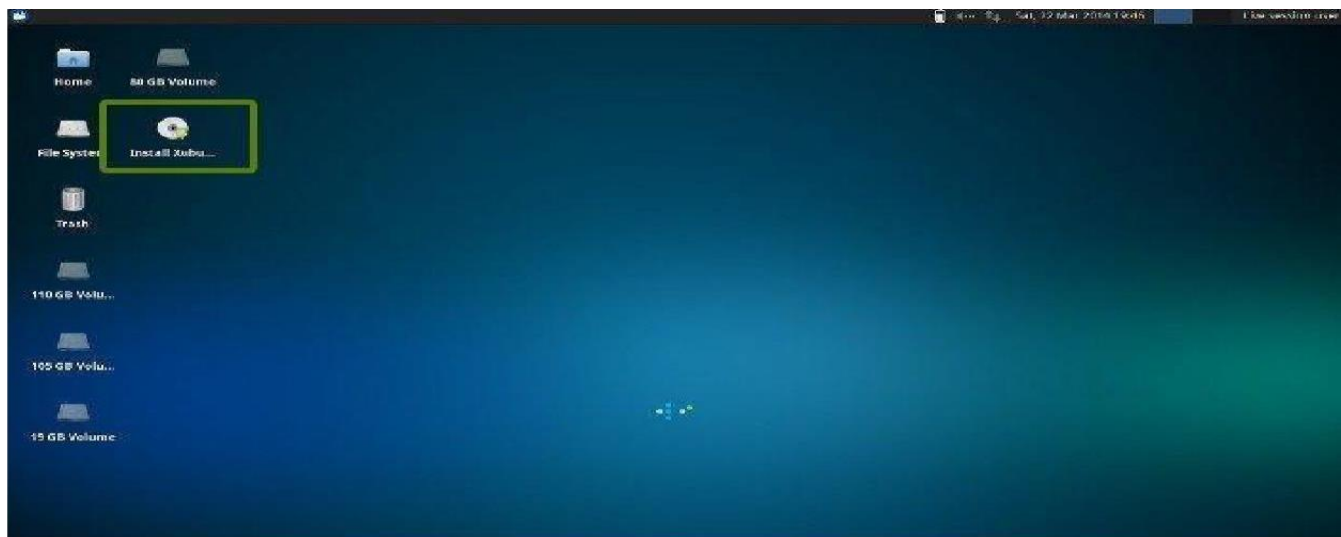
Download and create a live USB or DVD. In Windows, you can use tools like “universal USB installer “ to create a live Xubuntu USB

Step 2: Boot in to live USB

Plug the live USB or disk in to the computer and restart the computer. While booting the computer press F10 or F12 function key (defers from computer to computer) to go to the boot menu. Now, choose the option to boot from **USB** or **CD/DVD**.

Step 3: Start the installation

It will take some time to boot in to the live USB or disk. Once booted, you will be immediately provided with option to either try XUbuntu or install XUbuntu. Even if you choose to try, you can find the option to install on the desktop:

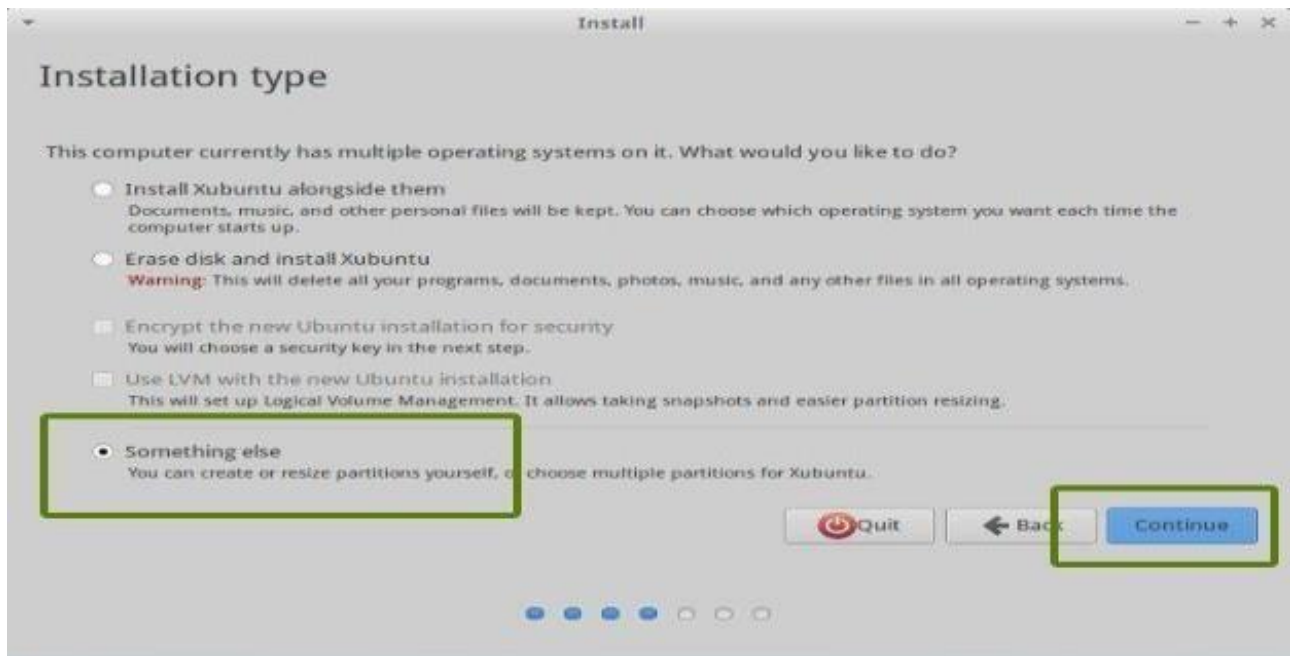


First few screens are pretty straight forward. Just choose press continue:

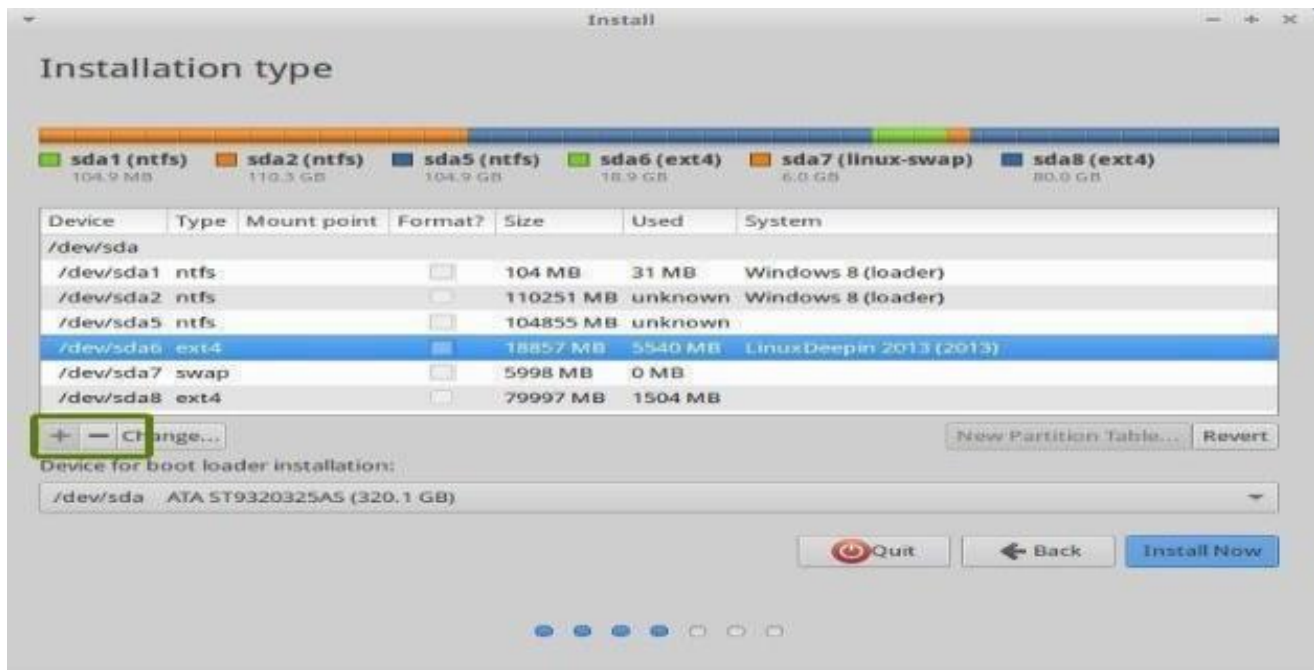


Step 4: Prepare the partition

This is the most important part of the whole dual boot installation. Where to install Xubuntu? Windows is already installed here, so, we'll prepare a new partition for Xubuntu. In the Installation Type window, choose **Something Else**:

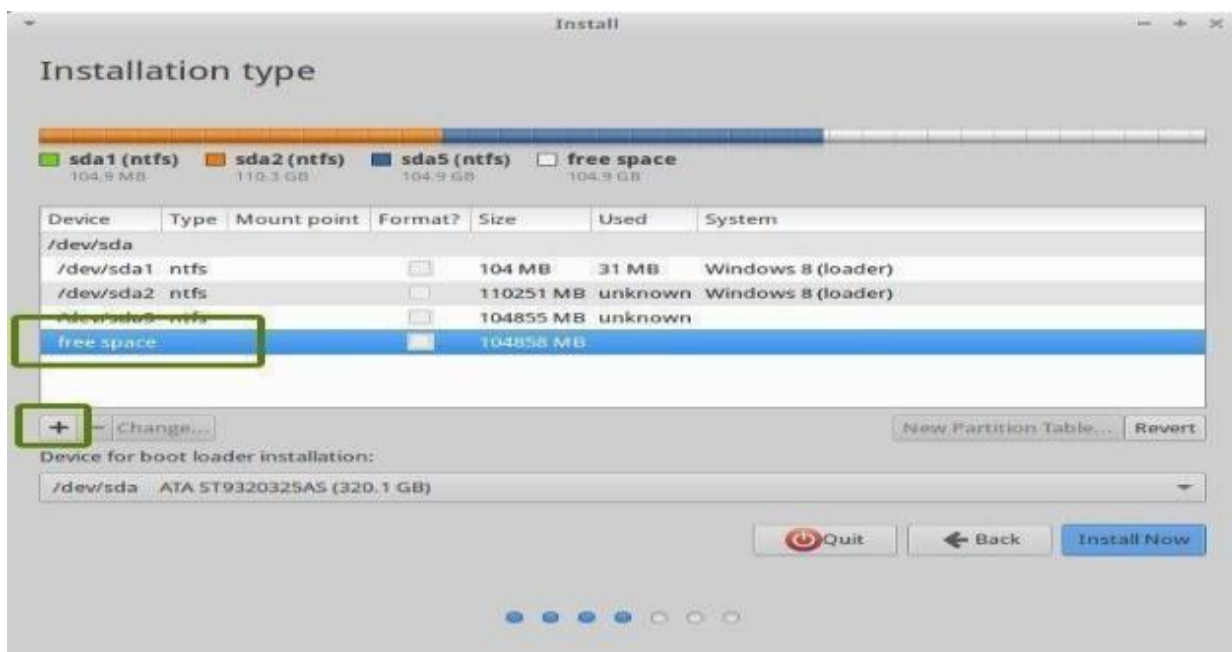


As you can see, it has 3 NTFS and some ext4 partitions. One of the NTFS partition consists of Windows installation. This should be untouched if you want to keep your Windows installation safe. If there is no “free space”, you need to delete a NTFS or existing ext4 partition and create some free space. This will delete all the data in that partition. Click on the desired partition and press the **–** to delete the partition.

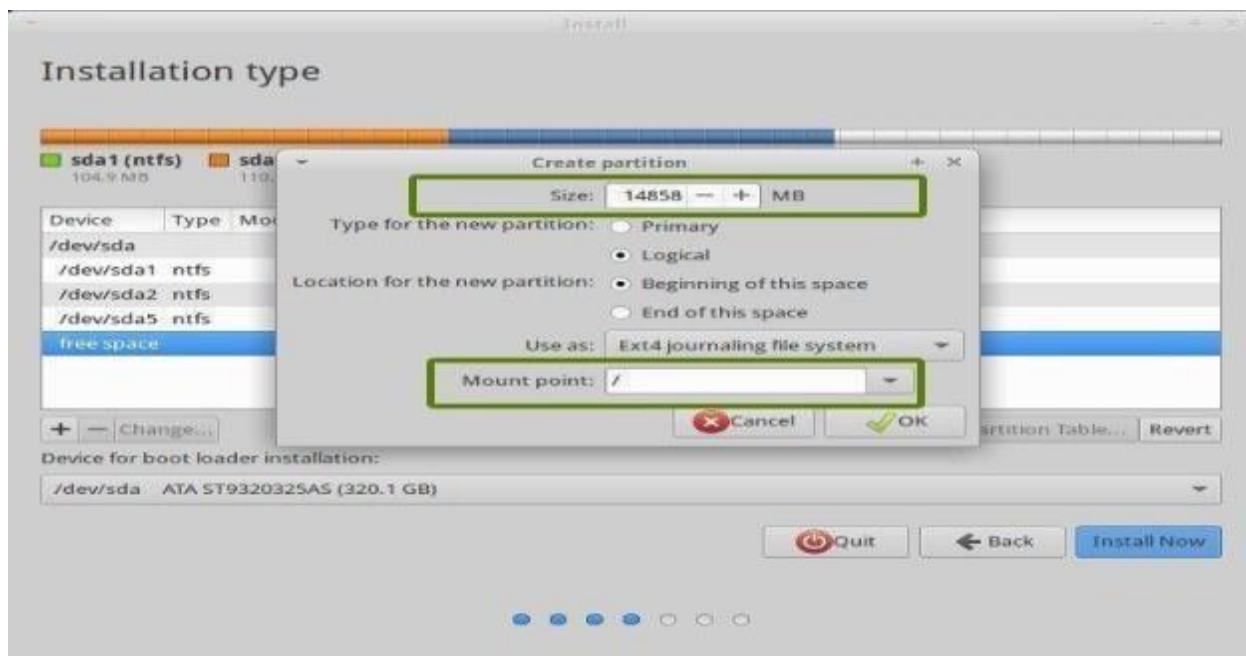


Step 5: Create root, swap and home

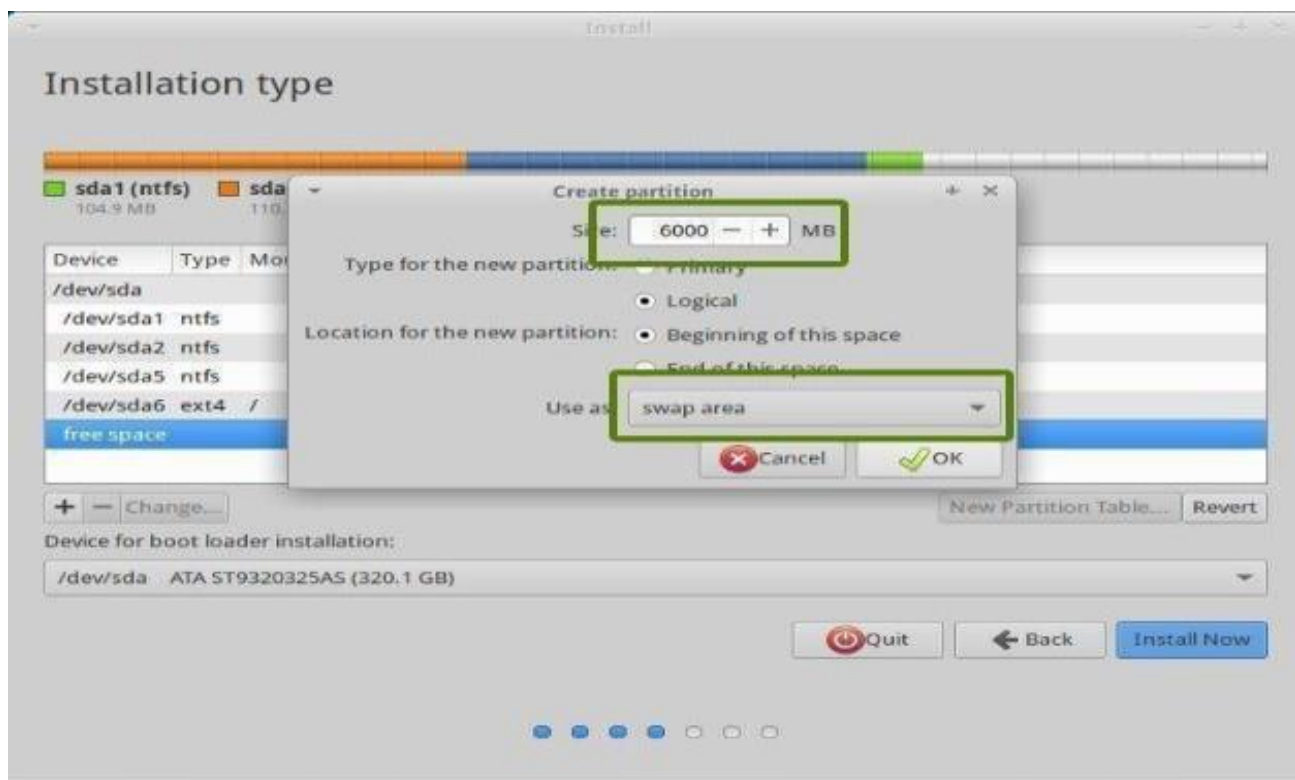
Once you have some free space on your hard drive, its time to install Ubuntu on it. Now, there are several ways to do it. We have to create minimum a Root, a Swap and a Home partition. Create a root partition first. Choose the free space available and click on **+**.



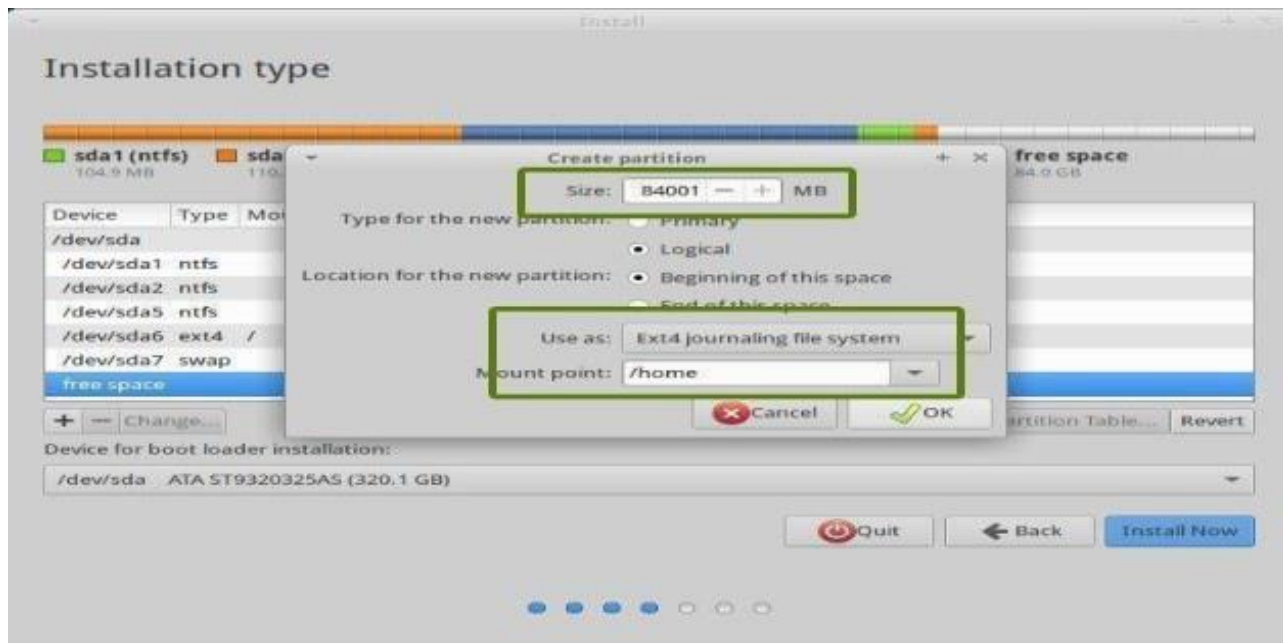
Here, choose the size of root directory, choose ext4 file system, and mount point as / (i.e. root):



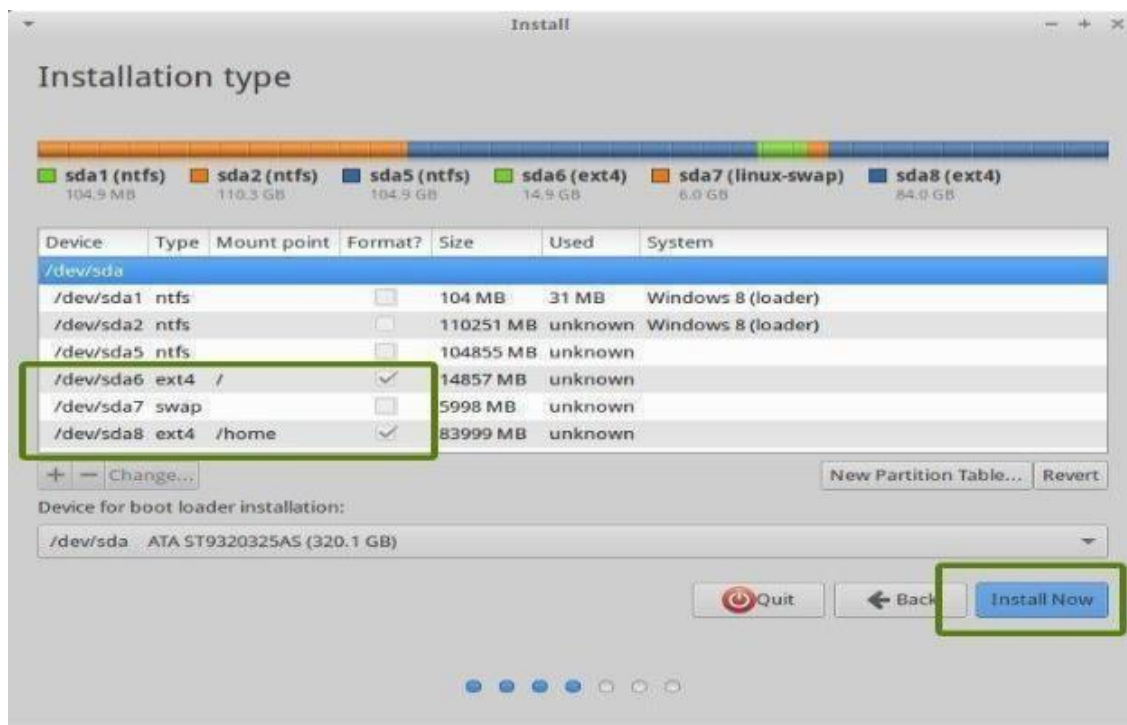
Next step is to create swap partition. It is advised by many that Swap should be double of your system's RAM size. You can choose the swap size accordingly.



The next step is to create Home. Try to allocate the maximum size to Home because this is where you'll be downloading and keeping the files.

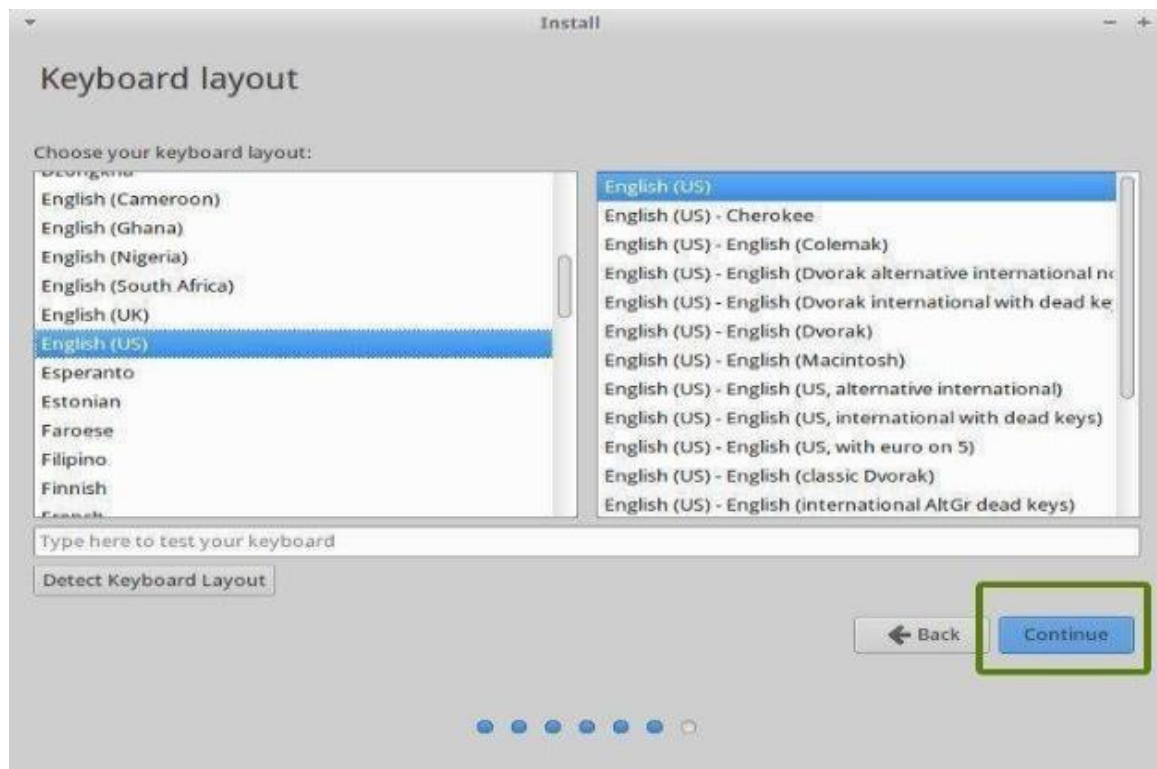
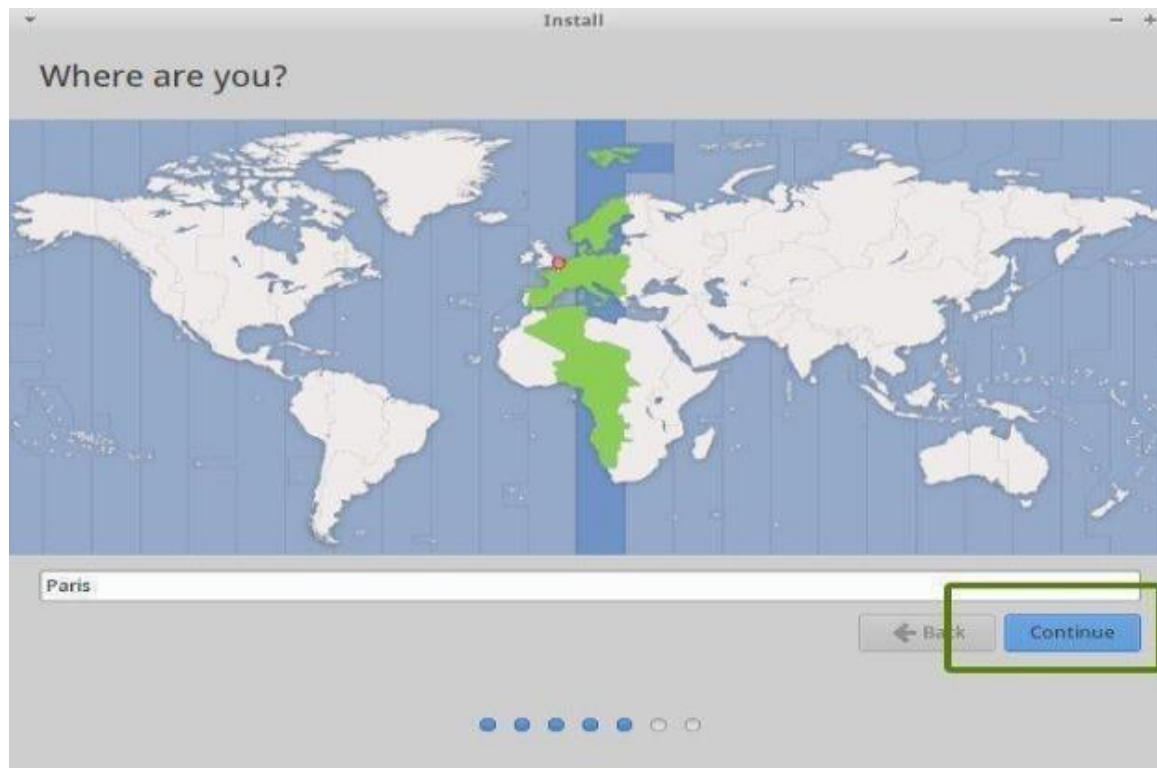


Once you have created Root, Swap and Home partitions, click on Install Now button.



Step 6: Follow the trivial instructions

If you successfully created the partitions as mentioned above, you will be taken through a number of screens to select options like keyboard layout, login credentials etc.



Install

Who are you?

Your name: ✓

Your computer's name: ✓
The name it uses when it talks to other computers.

Pick a username: ✓

Choose a password: **Good password**

Confirm your password: ✓

☐ Log in automatically
☒ Require my password to log in
☐ Encrypt my home folder

← Back **Continue**

• • • • • • • •

Install

Welcome to Xubuntu

»

Congratulations! You have chosen to install Xubuntu 13.10.

This exciting piece of software is brought to you free of any fees, for your use and any others you want to share it with.

Since you are installing this software using the Xubuntu live installation, feel free to examine things as it installs. After the installation, the desktop will look much the same as it does now.



Ubuntu

+



Xfce

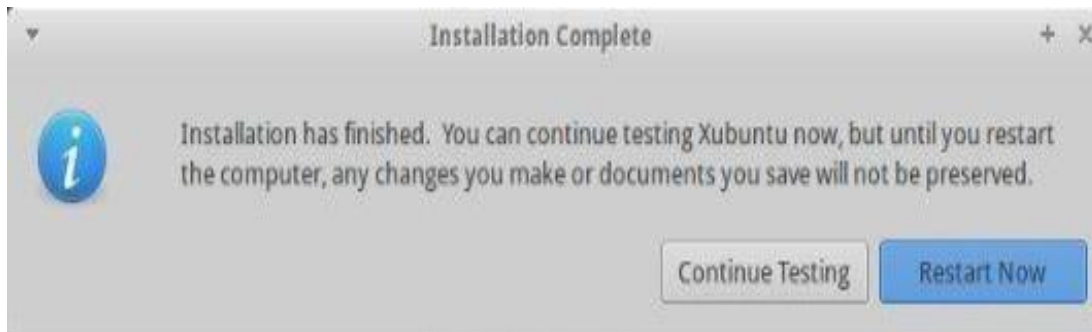
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Xubuntu

▶ Installing system Skip

Once the installation is over, you will be presented with the option to keep trying live version or to restart the system.



On next boot, you will see the option of XUbuntu in the grub screen.

Result

Familiarization of OS installation is done successfully

EXPERIMENT 2

STUDY OF TERMINAL BASED TEXT EDITOR VI & BASIC LINUX COMMANDS

a. Vi Editor commands

AIM

To study the various commands operated in Vi editor in LINUX.

DESCRIPTION

The Vi editor is a visual editor used to create and edit text, files, documents and programs. It displays the content of file on the screen and allows a user to add, delete or change part of text. There are two modes available in the

Vi editor, they are: -

- 1.Command mode
- 2.Input (or) insert mode

Starting Vi

Syntax:

\$vi<filename> (or)

\$vi

The command would open a display screen with 25 lines and with tilde (~) symbol at the start of each line. The first syntax would save the file in the filename mentioned and for then exit the file name must be mentioned at the end.

INSERTING AND READING COMMANDS

To move editor from command mode to edit mode, you have to press the <Esc> tag. For inserting and replacing the following commands are used.

1) <Esc>a command

This command is used to move the edit mode and start to append after the current character.

2) <Esc> A command

This command is also used to append the file, but this command append at the end of current line.

3) <Esc> i command

This command is used to insert the text before its current cursor position.

4) <Esc>I command

This command is used to insert at the beginning of the current file.

5) <Esc>o command

This command is inserting a blank line below the current line and allow insertion of contents.

6) <Esc>O command

This command is used to insert a blank line above &allow insertion of contents.

7) <Esc>r command

This command is to replace the particular character with the given character.

8) <Esc>R command

This command is used to replace the particular text with a given text.

9) <Esc>s command

This command replaces a single character with a group of character.

10) <Esc>S command

This command is used to replace a current line with group of characters.

DELETING THE TEXT FROM Vi

1) <Esc>x command

To delete a character to right of current cursor positions.

2) <Esc>X command

To delete a character to left of current cursor positions.

3) <Esc>dw command

This command is to delete a single word or number of word sorting of current cursor position.

4) <Esc>db command

This command is to delete a single word to the left of the current cursor position.

5) <Esc>dd command

This command is used to delete the current line(or) a number of lines below the current line.

6) <Esc>d\$ command

This command is used to delete the text from current cursor position to next character of current line.

SAVING AND QUITTING FROM Vi

1)<Esc>d\$ command

To save the given text present in the file.

2)<Esc>q! command

To quit the given text without saving.

3)<Esc>wq command

This command quits the Vi editor after saving the text in the mentioned file.

4)<Esc>x command

This command is same as “wq” command it save and quit.

5)<Esc>q command

This command would quit the window but it would ask for again to save the file.

b. BASIC LINUX COMMANDS

1. Pwd

Use the **pwd** command to find out the path of the current working directory (folder) you're in. The command will return an absolute (full) path, which is basically a path of all the directories that starts with a forward slash (/). An example of an absolute path is **/home/username**.

2. cd

To navigate through the Linux files and directories, use the **cd** . It requires either the full path or the name of the directory, depending on the current working directory that you're in. Let's say you're in **/home/username/Documents** and you want to go to **Photos**, a subdirectory of **Documents**.

To do so, simply type the following command: **cd Photos**. Another scenario is if you want to switch to a completely new directory, for example, **/home/username/Movies**. In this case, you have to type **cd** followed by the directory's absolute path: **cd /home/username/Movies**.

There are some shortcuts to help you navigate quickly:

- **cd ..** (with two dots) to move one directory up
- **cd** to go straight to the home folder
- **cd-** (with a hyphen) to move to your previous directory

On a side note, Linux's shell is case sensitive. So, you have to type the name's directory exactly as it is.

3. ls

The **ls** command is used to view the contents of a directory. By default, this command will display the contents of your current working directory. If you want to see the content of other directories, type **ls** and then the directory's path. For example, enter **ls /home/username/Documents** to view the content of **Documents**.

There are variations you can use with the **ls** command:

- **ls -R** will list all the files in the sub-directories as well
- **ls -a** will show the hidden files

- **ls -al** will list the files and directories with detailed information like the permissions, size, owner, etc.
- **ls -t** lists files sorted in the order of “last modified”
- **-r** option will reverse the natural sorting order. Usually used in combination with other switches such as **ls -tr**. This will reverse the time-wise listing.

4. cat

cat (short for concatenate) is one of the most frequently used commands in Linux. It is used to list the contents of a file on the standard output stdout . To run this command, type **cat** followed by the file’s name and its extension. For instance: **cat file.txt**.

Here are other ways to use the **cat** command:

- **cat > filename** creates a new file
- **cat filename1 filename2>filename3** joins two files (1 and 2) and stores the output of them in a new file (3)
- to convert a file to upper or lower case use, **cat filename | tr a-z A-Z >output.txt**

5. cp

Use the **cp** command to copy files from the current directory to a different directory. For instance, the command **cp scenery.jpg /home/username/Pictures** would create a copy of **scenery.jpg** (from your current directory) into the **Pictures** directory.

- **cp -i** will ask for user’s consent in case of a potential file overwrite.
- **cp -p** will preserve source files’ mode, ownership and timestamp.
- **cp -r** will copy directories recursively.
- **cp -u** copies files only if the destination file is not existing or the source file is newer than the destination file.

6. mv

The primary use of the **mv** command is to move files, although it can also be used to rename files. The arguments in **mv** are similar to the **cp** command. You need to type **mv**, the file’s name, and the destination’s directory. For example: **mv file.txt /home/username/Documents**. To rename files, the Linux is **mv oldname.ext newname.ext**

7. **mkdir**

Use **mkdir** command to make a new directory — if you type **mkdir Music** it will create a directory called **Music**.

There are extra **mkdir** commands as well:

- To generate a new directory inside another directory, use this Linux basic command **mkdir Music/Newfile**
- use the **p** (parents) option to create a directory in between two existing directories. For example, **mkdir -p Music/2020/Newfile** will create the new “2020” file.

8. **rmdir**

If you need to delete a directory, use the **rmdir** command. However, **rmdir** only allows you to delete empty directories.

9. **rm**

The **rm** command is used to delete directories and the contents within them. If you only want to delete the directory — as an alternative to **rmdir** — use **rm -r**.

Note: Be very careful with this command and double-check which directory you are in. This will delete everything and there is no undo.

10. **touch**

The **touch** command allows you to create a blank new file through the Linux command line. As an example, enter **touch /home/username/Documents/Web.html** to create an HTML file entitled **Web** under the **Documents** directory.

11. **grep**

Another basic Linux command that is undoubtedly helpful for everyday use is **grep**. It lets you search through all the text in a given file. To illustrate, **grep blue notepad.txt** will search for the word **blue** in the **notepad** file. Useful for command line use as well. Usually output of a previous command is piped into the **grep** command.

For example **ls -l | grep “kernel**

12. **sudo**

Short for “**SuperUser Do**”, this command enables you to perform tasks that require administrative or root permissions. You must have sufficient permissions to use this command.

13. **df**

Use **df** command to get a report on the system's disk space usage, shown in percentage and KBs. If you want to see the report in megabytes, type **df -m**.

14. **head**

The **head** command is used to view the first lines of any text file. By default, it will show the first ten lines, but you can change this number to your liking.

15. **tail**

This one has a similar function to the head command, but instead of showing the first lines, the **tail** command will display the last ten lines of a text file. For example, **tail -n filename.ext**.

16. **tar**

The **tar** command is the most used command to archive multiple files into a **tarball** — a common Linux file format that is similar to zip format, with compression being optional. This command is quite complex with a long list of functions such as adding new files into an existing archive, listing the content of an archive, extracting the content from an archive, and many more.

17. **chmod**

chmod is another Linux command, used to change the read, write, and execute permissions of files and directories. Read about permissions and how to manipulate them .

18. **chown**

In Linux, all files are owned by a specific user. The **chown** command enables you to change or transfer the ownership of a file to the specified username. For instance, **chown linuxuser2 file.ext** will make **linuxuser2** as the owner of the **file.ext**.

19. **ps**

Ps command will display all current processes along with their process ids (PID)

20. **ping**

Use the **ping** command to check your connectivity status to a server. For example, by simply entering **ping google.com**, the command will check whether you're able to connect to Google and also measure the response time.

21. **wget**

The Linux command line is super useful — you can even download files from the internet with the help of the **wget** command. To do so, simply type **wget** followed by the download link.

22. **uname**

The **uname** command, short for Unix Name, will print detailed information about your Linux system like the machine name, operating system, kernel, and so on.

23. **top**

As a terminal equivalent to Task Manager in Windows, the **top** command will display a list of running processes and how much CPU each process uses. It's very useful to monitor system resource usage, especially knowing which process needs to be terminated because it consumes too many resources.

24. **man**

Confused about the function of certain Linux commands? Don't worry, you can easily learn how to use them right from Linux's shell by using the **man** command. For instance, entering **man tail** will show the manual instruction of the tail command. Use the command: **man** to start learning about man utility.

25. **echo**

This command is used to move some data into a file. For example, if you want to add the text, "Hello, my name is John" into a file called name.txt, you would type **echo Hello, my name is John >> name.txt**

26. **zip, unzip**

Use the **zip** command to compress your files into a zip archive, and use the **unzip** command to extract the zipped files from a zip archive. (This program should be installed , some distributions may not have them. You can also look at gzip and bzip commands)

27. **hostname**

If you want to know the name of your host/network simply type **hostname**. Adding a **-I** to the end will display the IP address of your network.

Result

Study about Vi editor commands and Linux commands are successfully done

EXPERIMENT 3

STUDY OF FILE HIERARCHY IN COMMON LINUX DISTRIBUTION

It makes sense to explore the Linux file system from a terminal window, In fact, that is the name of the first tool you'll install to help you on the way: *tree*. If you are using Ubuntu or Debian, you can do:

sudo apt install tree

Once installed, stay in your terminal window and run *tree* like this: **\$ tree /**

The **/** in the instruction above refers to the *root* directory. The root directory is the one from which all other directories branch off from. When you run *tree* and tell it to start with **/**, you will see the whole directory tree, all directories and all the subdirectories in the whole system, with all their files, fly by.

If you have been using your system for some time, this may take a while, because, even if you haven't generated many files yourself, a Linux system and its apps are always logging, caching, and storing temporary files. The number of entries in the file system can grow quite quickly. Instead, try this:

tree -L 1 /

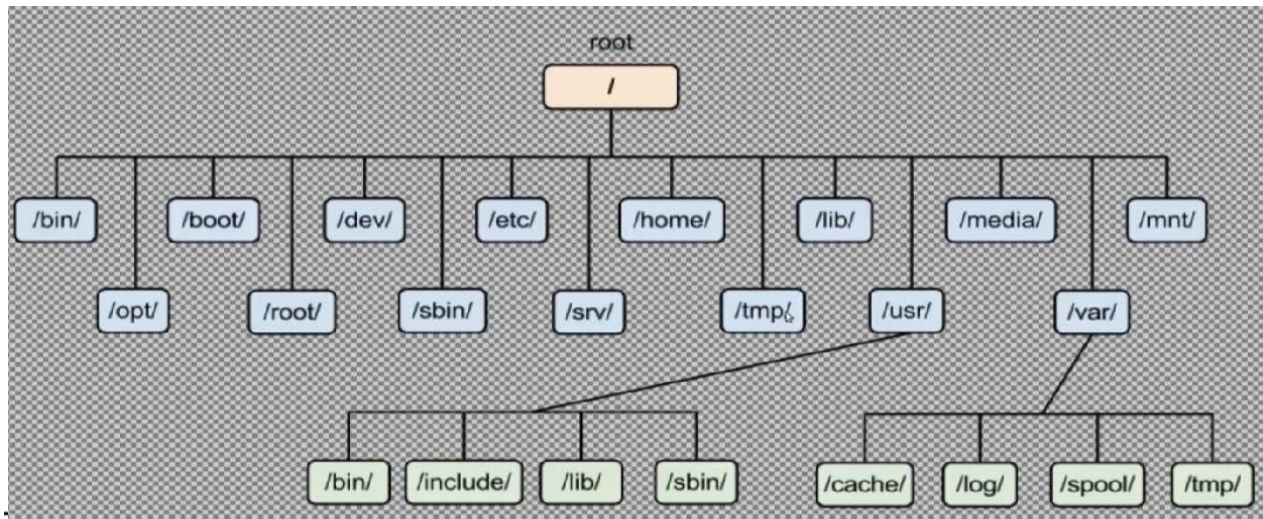
The instruction above can be translated as “*show me only the 1st Level of the directory tree starting at / (root)*”.

The **-L** option tells *tree* how many levels down you want to see.

Most Linux distributions will show you the same or a very similar layout to what you can see in the image above. This means that even if you feel confused now, master this, and you will have a handle on most, if not all, Linux installations in the whole wide world.

Now, let's look at what each directory is used for. While we go through each, you can peek at their contents using *ls*.

Directories



From top to bottom, the directories you are seeing are as follows.

/bin

/bin is the directory that contains *binaries*, that is, some of the applications and programs you can run. You will find the *ls* program mentioned above in this directory, as well as other basic tools for making and removing files and directories, moving them around, and so on. There are more *bin* directories in other parts of the file system tree, but we'll be talking about those in a minute.

/boot

The */boot* directory contains files required for starting your system. If you mess up one of the files in here, you may not be able to run your Linux and it is a pain to repair. On the other hand, don't worry too much about destroying your system by accident: you have to have superuser privileges to do that.

/dev

/dev contains *device* files. Many of these are generated at boot time or even on the fly. For example, if you plug in a new webcam or a USB pendrive into your machine, a new device entry will automatically pop up here.

/etc

/etc is the directory where names start to get confusing. */etc* gets its name from the earliest Unixes and it was literally "et cetera" because it was the dumping ground for system files administrators were not sure where else to put. Nowadays, it would be more appropriate to say that *etc* stands for "Everything to configure," as it

contains most, if not all system-wide configuration files. For example, the files that contain the name of your system, the users and their passwords, the names of machines on your network and when and where the partitions on your

/home

/home is where you will find your users' personal directories. In my case, under */home* there are two directories: */home/paul*, which contains all my stuff; and */home/guest*, in case anybody needs to borrow my computer.

/lib

/lib is where *libraries* live. Libraries are files containing code that your applications can use. They contain snippets of code that applications use to draw windows on your desktop, control peripherals, or send files to your hard disk. There are more *lib* directories scattered around the file system, but this one, the one hanging directly off of */* is special in that, among other things, it contains the all-important kernel modules. The kernel modules are drivers that make things like your video card, sound card, WiFi, printer, and so on, work.

/media

The */media* directory is where external storage will be automatically mounted when you plug it in and try to access it. As opposed to most of the other items on this list, */media* does not hail back to 1970s, mainly because inserting and detecting storage (pendrives, USB hard disks, SD cards, external SSDs, etc) on the fly, while a computer is running, is a relatively new thing.

/mnt

The */mnt* directory, however, is a bit of remnant from days gone by. This is where you would manually mount storage devices or partitions. It is not used very often nowadays.

/opt

The */opt* directory is often where software you compile (that is, you build yourself from source code and do not install from your distribution repositories) sometimes lands. Applications will end up in the */opt/bin* directory and libraries in the */opt/lib* directory.

A slight digression: another place where applications and libraries end up in is */usr/local*. When software gets installed here, there will also be */usr/local/bin* and */usr/local/lib* directories. What determines which software goes where is how the developers have configured the files that control the compilation and installation process

/proc

/proc, like */dev* is a virtual directory. It contains information about your computer, such as information about your CPU and the kernel your Linux system is running. As with */dev*, the files and directories are generated when your computer starts, or on the fly, as your system is running and things change.

/root

/root is the home directory of the superuser (also known as the “Administrator”) of the system. It is separate from the rest of the users’ home directories BECAUSE YOU ARE NOT MEANT TO TOUCH IT. Keep your own stuff in you own directories, people.

/run

/run is another new directory. System processes use it to store temporary data for their own nefarious reasons.

/sbin

/sbin is similar to */bin*, but it contains applications that only the superuser (hence the initial *s*) will need. You can use these applications with the *sudo* command that temporarily concedes you superuser powers on many distributions. */sbin* typically contains tools that can install stuff, delete stuff and format stuff. As you can imagine, some of these instructions are lethal if you use them improperly, so handle with care.

/usr

The */usr* directory was where users’ home directories were originally kept back in the early days of UNIX. However, now */home* is where users kept their stuff as we saw above. These days, */usr* contains a mish-mash of directories which in turn contain applications, libraries, documentation, wallpapers, icons and a long list of other stuff that need to be shared by applications and services. You will also find *bin*, *sbin* and *lib* directories in */usr*. What is the difference with their roothanging cousins? Not much nowadays. Originally, the */bin* directory (hanging off of root) would contain very basic commands, like *ls*, *mv* and *rm*; the kind of commands that would come preinstalled in all UNIX/Linux installations, the bare minimum to run and maintain a system. */usr/bin* on the other hand would contain stuff the users would install and run to use the system as a work station, things like word processors, web browsers, and other apps.

But many modern Linux distributions just put everything into */usr/bin* and have */bin* point to */usr/bin* just in case erasing it completely would break something. So, while Debian, Ubuntu and Mint still keep */bin* and */usr/bin* (and */sbin* and */usr/sbin*) separate; others, like Arch and its derivatives just have one “real” directory forbinaries, */usr/bin*, and the rest or **bins* are “fake” directories that point to */usr/bin*.

/srv

The */srv* directory contains data for servers. If you are running a web server from your Linux box, your HTML files for your sites would go into */srv/http* (or */srv/www*). If you were running an FTP server, your files would go into */srv/ftp*.

/sys

/sys is another virtual directory like */proc* and */dev* and also contains information from devices connected to your computer. In some cases you can also manipulate those devices. I can, for example, change the brightness of the screen of my laptop by modifying the value stored in the */sys/devices/pci0000:00/0000:00:02.0/drm/card1/card1-eDP-1/intel_backlight/brightness* file. But to do that you have to become superuser. The reason for that is, as with so many other virtual directories, messing with the contents and files in */sys* can be dangerous and you can trash your system. DO NOT TOUCH until you are sure you know what you are doing.

/tmp

/tmp contains temporary files, usually placed there by applications that you are running. The files and directories often (not always) contain data that an application doesn't need right now, but may need later on. You can also use */tmp* to store your own temporary files — */tmp* is one of the few directories hanging off / that you can actually interact with without becoming superuser.

/var

/var was originally given its name because its contents was deemed *variable*, in that it changed frequently. Today it is a bit of a misnomer because there are many other directories that also contain data that changes frequently, especially the virtual directories we saw above.

Be that as it may, */var* contains things like logs in the */var/log* subdirectories. Logs are files that register events that happen on the system. If something fails in the kernel, it will be logged in a file in */var/log*; if someone tries to break into your computer from outside, your firewall will also log the attempt here. It also contains *spools* for tasks. These “tasks” can be the jobs you send to a shared printer when you have to wait because another user is printing a long document, or mail that is waiting to be delivered to users on the system.

Your system may have some more directories we haven't mentioned above. In the screenshot, for example, there is a */snap* directory. That's because the shot was captured on an Ubuntu system. Ubuntu has recently incorporated snap packages as a way of distributing software. The */snap* directory contains all the files and the software installed from snaps. */var/log/syslog* contains lot of system related logfiles. To explore the filesystem yourself, use the *cd* command:

cd will take you to the directory of your choice (*cd* stands for *change directory*). If you get confused,

pwd

will always tell you where you (*pwd* stands for *print working directory*). Also, ***cd*** with no options or parameters, will take you back to your own home directory, where things are safe and cosy. Finally, ***cd ..*** will take you up one level, getting you one level closer to the */* root directory. If you are in */usr/share/wallpapers* and run *cd ..*, you will move up to */usr/share*.

ls to list the contents of the directory you are in right now. And, of you always have *tree* to get an overview of what lays within a directory. Try it on */usr/share* — there is a lot of interesting stuff in there.

Result

Study of File Hierarchy in common linux distribution is done successfully.

EXPERIMENT 4

SHELL SCRIPTING PROGRAMS

1. ARITHMETIC OPERATION

AIM:

Write a shell program to perform arithmetic operation.

ALGORITHM:

1. Start
2. Initialize two variables x and y with values:
 - x = 20
 - y = 10
3. Calculate the sum = x + y
4. Calculate the difference = x - y
5. Calculate the product = x * y
6. Calculate the quotient = x / y
7. Calculate the remainder = x % y
8. Display all results using echo:
 - Print Sum
 - Print Difference
 - Print Product
 - Print Quotient
 - Print Remainder
9. End

PROGRAM

```
x=20
y=10
sum=$(( $x + $y ))
difference=$(( $x - $y ))
product=$(( $x * $y ))
quotient=$(( $x / $y ))
remainder=$(( $x % $y ))
```

```
echo "Sum=$sum"  
echo "Difference=$difference"  
echo "Product=$product"  
echo "Quotient=$quotient"  
echo "Remainder=$remainder"
```

OUTPUT

```
root@LAPTOP-1B2435NS:~/S2MCA# chmod +x arithmetic.sh  
root@LAPTOP-1B2435NS:~/S2MCA# ./arithmetic.sh  
Sum=30  
Difference=10  
Product=200  
Quotient=2  
Remainder=0  
root@LAPTOP-1B2435NS:~/S2MCA#
```

RESULT:

Thus, the program has been executed successfully.

2. AVERAGE OF NUMBERS

AIM:

Write a shell script to find average of three numbers.

ALGORITHM:

1. Start
2. Prompt the user to enter the first number
3. Read the first number and store it in variable x
4. Prompt the user to enter the second number
5. Read the second number and store it in variable y
6. Prompt the user to enter the third number
7. Read the third number and store it in variable z
8. Calculate the sum:
$$\text{sum} = x + y + z$$
9. Calculate the average:
$$\text{average} = \text{sum} / 3$$
10. Display the average using echo
11. End

PROGRAM

```
echo "Enter the first number:"
read x
echo "Enter the second number:"
read y
echo "Enter the third number:"
read z
sum=$(( $x + $y+$z ))
average=$(( $sum / 3 ))
echo "The average=$average"
```

OUTPUT

```
root@DESKTOP-PNM9727:~/S2MCA# ./average.sh
Enter the first number
10
Enter the second number
20
Enter the third number
30
The average = 10
```

RESULT:

Thus, the program has been executed successfully.

3. SWAP TWO NUMBERS

AIM:

Write a shell script for swapping two numbers.

ALGORITHM

1. Start
2. Prompt the user to enter two numbers
3. Read the first number and store it in variable x
4. Read the second number and store it in variable y
5. Display the values of x and y before swapping
6. Store the value of x in a temporary variable temp
7. Assign the value of y to x
8. Assign the value of temp to y
9. Display the values of x and y after swapping
10. End

PROGRAM

```
echo "Enter two numbers:"
read x
read y
echo -e "Before swapping:\nx=$x\ty=$y"
temp=$x
x=$y
y=$temp
echo -e "After swapping:\nx=$x\ty=$y"
```

OUTPUT

```
root@DESKTOP-PNM9727:~/S2MCA# ./swap.sh
Enter two numbers:
10
25
Before Swapping:
x=10    y=25
After Swapping:
x=25    y=10
```

RESULT:

Thus, the program has been executed successfully.

4. REVERSE OF A NUMBER

AIM

Write a Shell Script to Reverse a 5-Digit Number.

ALGORITHM

1. Start
2. Prompt the user to enter a 5-digit number
3. Read the number and store it in variable x
4. Initialize rev = 0
5. Repeat the following steps while x > 0:
6. a. Calculate remainder: $\text{rem} = x \% 10$
7. b. Update reverse number: $\text{rev} = \text{rev} * 10 + \text{rem}$
8. c. Remove the last digit from x: $x = x / 10$
9. Display the reversed number using echo
10. End

PROGRAM

```
echo "Enter a 5-digit number:"
read x
rev=0
while [ $x -gt 0 ]
do
    rem=$(( $x % 10 ))
    rev=$(( $rev * 10 + $rem ))
    x=$(( $x / 10 ))
done
echo "Reversed number is: $rev"
```

OUTPUT

```
root@DESKTOP-PNM9727:~/S2MCA# ./reverse.sh
Enter a 5-digit number:
12345
Reversed number is:54321
root@DESKTOP-PNM9727:~/S2MCA#
```

RESULT:

Thus, the program has been executed successfully.

5. LARGEST AMONG THREE NUMBERS

AIM

Write a shell script to find the largest among three numbers.

ALGORITHM

1. Start
2. Prompt the user to enter three numbers
3. Read the first number and store it in variable a
4. Read the second number and store it in variable b
5. Read the third number and store it in variable c
6. Check if a is greater than both b and c:
 - If true, print "a is greater"
7. Else, check if b is greater than both a and c:
 - If true, print "b is greater"
8. Else:
 - Print "c is greater"
9. End

PROGRAM

```
echo "Enter three numbers:"
read a
read b
read c
if [[ $a -gt $b && $a -gt $c ]];
then
    echo "$a is greater"
elif [[ $b -gt $a && $b -gt $c ]];
then
    echo "$b is greater"
else
    echo "$c is greater"
fi
```

OUTPUT

```
root@DESKTOP-PNM9727:~/S2MCA# ./largest.sh
Enter three number:
10
20
30
30 is greater
root@DESKTOP-PNM9727:~/S2MCA#
```

RESULT:

Thus, the program has been executed successfully.

6. SUM OF DIGITS

AIM

Write a shell script to find the sum of digits of a number.

ALGORITHM

1. Start
2. Prompt the user to enter a 5-digit number
3. Read the number and store it in variable x
4. Initialize sums = 0
5. Repeat the following steps while $x > 0$:
 - a. Calculate the last digit: $\text{rem} = x \% 10$
 - b. Add rem to sums: $\text{sums} = \text{sums} + \text{rem}$
 - c. Remove the last digit from x: $x = x / 10$
6. Display the sum of digits using echo
7. End

PROGRAM

```
echo "Enter a number:"
read x
sums=0
while [ $x -gt 0 ]
do
    rem=$(( $x % 10 ))
    sums=$(( $sums + $rem ))
    x=$(( $x / 10 ))
done
echo "The sum of digits is: $sums"
```

OUTPUT

```
root@DESKTOP-PNM9727:~/S2MCA# ./sumofdigits.sh
Enter a number:
123
The sum of digits is:6
root@DESKTOP-PNM9727:~/S2MCA#
```

RESULT:

Thus, the program has been executed successfully.

7. SUM OF N NUMBERS

AIM

Find the Sum of First N Natural Numbers in Shell Script

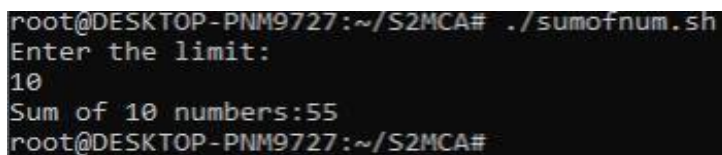
ALGORITHM

1. Start
2. Prompt the user to enter the limit (n)
3. Read the value and store it in variable n
4. Initialize sum = 0
5. Repeat the following steps from i = 1 to n:
 - a. Add i to sum: sum = sum + i
7. Display the final sum using echo
8. End

PROGRAM

```
echo "Enter the limit:"
read n
sum=0
for (( i=1; i <= n; i++ ))
do
sum=$(( sum + i ))
done
echo "Sum of $n numbers :$sum"
```

OUTPUT



```
root@DESKTOP-PNM9727:~/S2MCA# ./sumofnum.sh
Enter the limit:
10
Sum of 10 numbers:55
root@DESKTOP-PNM9727:~/S2MCA#
```

RESULT:

Thus, the program has been executed successfully.

8. EVEN OR ODD

AIM

Write a Shell Script to Print Even and Odd Numbers up to a Given Limit.

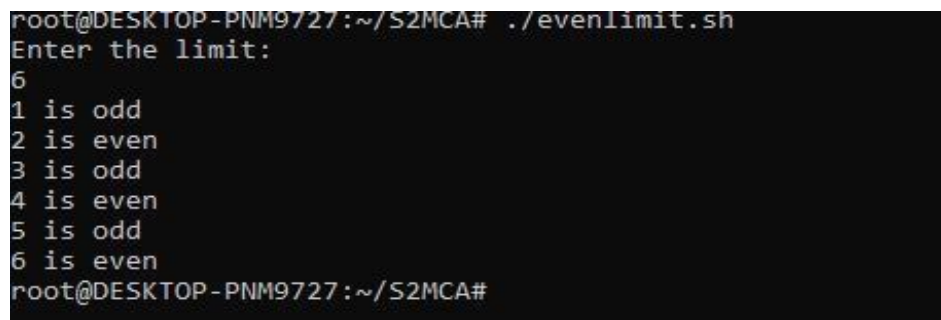
ALGORITHM

1. Start
2. Prompt the user to enter the limit
3. Read the value and store it in variable x
4. Repeat the following steps from i = 1 to x:
 - a. Check if $i \% 2 == 0$
 - If true, print "i is even"
 - Else, print "i is odd"
5. End

PROGRAM

```
echo "Enter the limit:"
read x
for (( i = 1; i <= x; i++ )); do
if [  $$(i \% 2)$  -eq 0 ]; then
echo "$i is even"
else
echo "$i is odd"
fi
done
```

OUTPUT



```
root@DESKTOP-PNM9727:~/S2MCA# ./evenlimit.sh
Enter the limit:
6
1 is odd
2 is even
3 is odd
4 is even
5 is odd
6 is even
root@DESKTOP-PNM9727:~/S2MCA#
```

RESULT:

Thus, the program has been executed successfully.

9. FACTORIAL OF N NUMBERS

AIM

Find the Factorial of a Number in Shell Script

ALGORITHM

1. Start
2. Prompt the user to enter a number
3. Read the number and store it in variable x
4. Initialize fact = 1
5. Repeat the following steps from i = 1 to x:
 - a. Multiply fact by i: fact = fact * i
 - b. Display the factorial of i
6. End

PROGRAM

```
echo "Enter a number:"
read x
fact=1
for (( i=1; i <= x; i++ ));
do
fact=$(( fact * i ))
echo "Factorial of $i is $fact"
done
```

OUTPUT

```
root@DESKTOP-PNM9727:~/S2MCA# ./factorial.sh
Enter the limit:
6
Factorial of 1 is 1
Factorial of 2 is 2
Factorial of 3 is 6
Factorial of 4 is 24
Factorial of 5 is 120
Factorial of 6 is 720
root@DESKTOP-PNM9727:~/S2MCA#
```

RESULT:

Thus, the program has been executed successfully.

10. READ A FILE

AIM

Write a shell script to Read and Display a File if It Exists in Shell Script

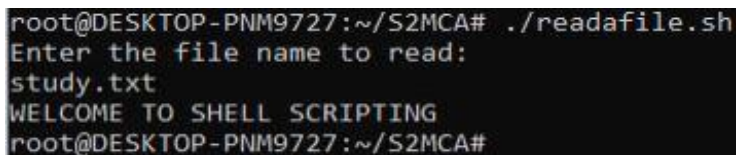
ALGORITHM

1. Start
2. Prompt the user to enter the file name to read
3. Read the file name and store it in variable filename
4. Check if the file exists using -f:
 - a. If true, display the contents of the file using cat
 - b. Else, display "file does not exist"
5. End

PROGRAM

```
echo "Enter the file name to read:"
read filename
if [[ -f "$filename" ]];
then
cat "$filename"
else
echo "file does not exist"
fi
```

OUTPUT



```
root@DESKTOP-PNM9727:~/S2MCA# ./readfile.sh
Enter the file name to read:
study.txt
WELCOME TO SHELL SCRIPTING
root@DESKTOP-PNM9727:~/S2MCA#
```

RESULT:

Thus, the program has been executed successfully.

EXPERIMENT 5

LAMP INSTALLATION

Open terminal and do the following steps:

Update your system

```
sudo apt update
```

Install Apache using apt:

```
sudo apt install apache2 -y
```

Install the Apache package:

```
sudo apt install apache2 -y
```

```
marko@test-main:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
The following NEW packages will be installed:
  apache2
0 upgraded, 1 newly installed, 0 to remove and 11 not upgraded.
Need to get 95.6 kB of archives.
After this operation, 543 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu focal-updates/main amd64 apache2 amd64 2.4.41-4ubuntu3.12 [95.6 kB]
Fetched 95.6 kB in 1s (119 kB/s)
Selecting previously unselected package apache2.
(Reading database ... 174557 files and directories currently installed.)
Preparing to unpack .../apache2_2.4.41-4ubuntu3.12_amd64.deb ...
Unpacking apache2 (2.4.41-4ubuntu3.12) ...
Setting up apache2 (2.4.41-4ubuntu3.12) ...
Processing triggers for systemd (245.4-4ubuntu3.18) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ufw (0.36-6ubuntu1) ...
marko@test-main:~$
```

Check if Apache installed correctly by checking the Apache service status:

```
sudo service apache2 status
```

The service shows as **running** in the output:

```
marko@test-main:~$ sudo service apache2 status
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enab
   Active: active (running) since Thu 2022-10-13 05:26:21 EDT; 4min 53s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 15228 (apache2)
      Tasks: 6 (limit: 11573)
     Memory: 9.4M
    CGroup: /system.slice/apache2.service
            └─15228 /usr/sbin/apache2 -k start
              └─15230 /usr/sbin/apache2 -k start
                └─15231 /usr/sbin/apache2 -k start
                  └─15233 /usr/sbin/apache2 -k start
                    └─15234 /usr/sbin/apache2 -k start
                      └─15235 /usr/sbin/apache2 -k start

Oct 13 05:26:21 test-main systemd[1]: Starting The Apache HTTP Server...
Oct 13 05:26:21 test-main apachectl[15227]: AH00558: apache2: Could not reliably deter
Oct 13 05:26:21 test-main systemd[1]: Started The Apache HTTP Server.
lines 1-18/18 (END)
```

Exit the status screen by pressing **Ctrl + C** on the keyboard.

Next, make sure that the UFW firewall contains the Apache profiles by typing in the following command:

sudo ufw app list

```
marko@test-main:~$ sudo ufw app list
Available applications:
  Apache
  Apache Full
  Apache Secure
  CUPS
marko@test-main:~$
```

Ensure the **Apache Full** profile allows the traffic on ports **80** and **443** by running the command:

sudo ufw app info "Apache Full"

```
marko@test-main:~$ sudo ufw app info "Apache Full"
Profile: Apache Full
Title: Web Server (HTTP,HTTPS)
Description: Apache v2 is the next generation of the omnipresent Apache web
server.

Ports:
  80,443/tcp
marko@test-main:~$
```

To confirm that Apache is running, enter the IP address of your server in the address bar of an internet browser and press **ENTER**.

The test Apache web server page should display as below.



Apache2 Ubuntu Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

Install MySQL and create a Database

```
sudo apt install mysql-server -y
```

```
Setting up mysql-server-8.0 (8.0.30-0ubuntu0.20.04.2) ...
update-alternatives: using /etc/mysql/mysql.cnf to provide /etc/mysql/my.cnf (my.cnf) in auto mode
mysqld will log errors to /var/log/mysql/error.log
mysqld is running as pid 19563
Setting up mysql-server (8.0.30-0ubuntu0.20.04.2) ...
Processing triggers for systemd (245.4-4ubuntu3.18) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
marko@test-main:~$
```

Install PHP

```
sudo apt install php libapache2-mod-php php-mysql -y
```

```
Setting up php7.4-mysql (7.4.3-4ubuntu2.13) ...
Creating config file /etc/php/7.4/mods-available/mysqlnd.ini with new version
Creating config file /etc/php/7.4/mods-available/mysqli.ini with new version
Creating config file /etc/php/7.4/mods-available/pdo_mysql.ini with new version
Setting up libapache2-mod-php (2:7.4+75) ...
Setting up php-mysql (2:7.4+75) ...
Processing triggers for libapache2-mod-php7.4 (7.4.3-4ubuntu2.13) ...
Processing triggers for php7.4-cli (7.4.3-4ubuntu2.13) ...
marko@test-main:~$
```

Modify the way Apache serves files by opening the ***dir.conf*** file in a text editor with root privileges:

```
sudo nano /etc/apache2/mods-enabled/dir.conf
```

The configuration file looks like in the example below:

```
GNU nano 4.8 /etc/apache2/mods-enabled/dir.conf
<IfModule mod_dir.c>
    DirectoryIndex index.html index.cgi index.pl index.php index.xhtml index.htm
</IfModule>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet
```

By default, Apache first looks for an **index.html** file card.

Edit the list so that the **index.php** file is in the first position:

```
GNU nano 4.8 /etc/apache2/mods-enabled/dir.conf Modified
<IfModule mod_dir.c>
    DirectoryIndex index.php index.html index.cgi index.pl index.xhtml index.htm
</IfModule>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet
```

Press **CTRL + X** to save and close the file. Press **y** and **ENTER** to confirm.

The command pipes the results of the **apt-cache** search into **less** to simplify viewing the output.

Restart Apache

```
sudo systemctl restart apache2
```

If the command executes correctly, it returns no output.

Test PHP Processing on Web Server

```
sudo nano /var/www/html/info.php
```

Inside the file, type the PHP code:

```
<?php
phpinfo ();
?>
```

```
GNU nano 4.8 /var/www/html/info.php Modified
<?php
phpinfo ();
?>
```

Press **CTRL + X** to save and close the file. Press **y** and **ENTER** to confirm.

Open an internet browser and type the following address:

```
[server-ip-address]/info.php
```

Alternatively, type:

```
localhost/info.php
```

The output should display the details of the LAMP stack, as seen in the image below:

← → ↺ localhost/info.php 80% ☆ ☑ ☰	
PHP Version 7.4.3 php	
System	Linux test-main 5.15.0-48-generic #54~20.04.1-Ubuntu SMP Thu Sep 1 16:17:26 UTC 2022 x86_64
Build Date	Aug 17 2022 13:29:56
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/7.4/apache2
Loaded Configuration File	/etc/php/7.4/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/7.4/apache2/conf.d
Additional .ini files parsed	/etc/php/7.4/apache2/conf.d/10-mysqld.ini, /etc/php/7.4/apache2/conf.d/10-opcache.ini, /etc/php/7.4/apache2/conf.d/10-pdo.ini, /etc/php/7.4/apache2/conf.d/20-calendar.ini, /etc/php/7.4/apache2/conf.d/20-ctype.ini, /etc/php/7.4/apache2/conf.d/20-exif.ini, /etc/php/7.4/apache2/conf.d/20-ffi.ini, /etc/php/7.4/apache2/conf.d/20-fileinfo.ini, /etc/php/7.4/apache2/conf.d/20-ftp.ini, /etc/php/7.4/apache2/conf.d/20-gettext.ini, /etc/php/7.4/apache2/conf.d/20-iconv.ini, /etc/php/7.4/apache2/conf.d/20-json.ini, /etc/php/7.4/apache2/conf.d/20-mysqli.ini, /etc/php/7.4/apache2/conf.d/20-pdo_mysql.ini, /etc/php/7.4/apache2/conf.d/20-phar.ini, /etc/php/7.4/apache2/conf.d/20-posix.ini, /etc/php/7.4/apache2/conf.d/20-readline.ini, /etc/php/7.4/apache2/conf.d/20-shmop.ini, /etc/php/7.4/apache2/conf.d/20-sockets.ini, /etc/php/7.4/apache2/conf.d/20-sysvmsg.ini, /etc/php/7.4/apache2/conf.d/20-sysvsem.ini, /etc/php/7.4/apache2/conf.d/20-sysvshm.ini, /etc/php/7.4/apache2/conf.d/20-tokenizer.ini
PHP API	20190902
PHP Extension	20190902
Zend Extension	320190902
Zend Extension Build	API320190902.NTS
PHP Extension Build	API20190902.NTS
Debug Build	no

Result

Linux Installation is successfully done

EXPERIMENT 6

NETWORK COMMANDS

1. ifconfig

Linux ifconfig stands for interface configurator. It is one of the most basic commands used in network inspection. ifconfig is used to initialize an interface, configure it with an IP address, and enable or disable it. It is also used to display the route and the network interface.

Basic information displayed upon using ifconfig are:

1. IP address
2. MAC address
3. MTU (Maximum Transmission Unit)

Syntax: ifconfig

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.21.64.206 netmask 255.255.240.0 broadcast 172.21.79.255
    inet6 fe80::215:5dff:fe54:be70 prefixlen 64 scopeid 0x20<link>
    ether 00:15:5d:54:be:70 txqueuelen 1000 (Ethernet)
    RX packets 42 bytes 4647 (4.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 15 bytes 1146 (1.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. traceroute

Linux traceroute is one of the most useful commands in networking. It is used to troubleshoot the network. It detects the delay and determines the pathway to your target.

Syntax

traceroute <destination>

Example: \$ traceroute google.com

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# traceroute google.com
traceroute to google.com (142.250.77.142), 30 hops max, 60 byte packets
 1  DESKTOP-PNM9727.mshome.net (172.21.64.1)  4.984 ms  4.387 ms  4.335 ms
 2  192.168.1.1 (192.168.1.1)  3.925 ms  5.245 ms  5.105 ms
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

3. tracepath

Linux tracepath is similar to traceroute command. It is used to detect network delays. However, it doesn't require root privileges.

It is installed in Ubuntu by default.

It traces the route to the specified destination and identifies each hop in it. If your network is weak, it recognizes the point where the network is weak.

Syntax

```
tracepath <destination>
```

Example: tracepath www.google.com

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# tracepath google.com
 1?: [LOCALHOST] pmtu 1500
 1:  DESKTOP-PNM9727.mshome.net 0.647ms
 1:  DESKTOP-PNM9727.mshome.net 0.346ms
 2:  192.168.1.1 5.700ms
 3:  192.168.1.1 4.216ms pmtu 1480
 3:  no reply
 4:  no reply
 5:  no reply
 6:  no reply
 7:  no reply
 8:  no reply
 9:  no reply
10:  no reply
11:  no reply
12:  no reply
13:  no reply
14:  no reply
15:  no reply
16:  no reply
17:  no reply
18:  no reply
19:  no reply
20:  no reply
21:  no reply
22:  no reply
23:  no reply
24:  no reply
25:  no reply
26:  no reply
27:  no reply
28:  no reply
29:  no reply
30:  no reply
Too many hops: pmtu 1480
Resume: pmtu 1480
```

4. ping

Linux ping is one of the most used network troubleshooting commands. It basically checks for the network connectivity between two nodes. ping stands for Packet INternet Groper. The ping command sends the ICMP echo request to check the network connectivity. It keeps executing until it is interrupted.

Use **Ctrl+C Key** to interrupt the execution.

Syntax:

```
ping <destination>
```

Example: \$ ping google.com

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# ping google.com
PING google.com (142.250.206.110) 56(84) bytes of data.
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=1 ttl=112 time=85.5 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=2 ttl=112 time=111 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=3 ttl=112 time=71.9 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=4 ttl=112 time=76.3 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=7 ttl=112 time=136 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=8 ttl=112 time=77.9 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=9 ttl=112 time=76.7 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=10 ttl=112 time=76.4 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=11 ttl=112 time=72.9 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=12 ttl=112 time=146 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=13 ttl=112 time=94.2 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=14 ttl=112 time=86.0 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=15 ttl=112 time=73.1 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=16 ttl=112 time=83.5 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=17 ttl=112 time=95.3 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=18 ttl=112 time=86.3 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=19 ttl=112 time=72.3 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=20 ttl=112 time=72.9 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=21 ttl=112 time=133 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=22 ttl=112 time=89.3 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=23 ttl=112 time=82.7 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=24 ttl=112 time=76.6 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=25 ttl=112 time=130 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=26 ttl=112 time=75.5 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=27 ttl=112 time=78.0 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=28 ttl=112 time=73.2 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=29 ttl=112 time=89.7 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=30 ttl=112 time=75.2 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=31 ttl=112 time=72.0 ms
64 bytes from del11s20-in-f14.1e100.net (142.250.206.110): icmp_seq=32 ttl=112 time=75.0 ms
^Z
[4]+  Stopped                  ping google.com
```

The ping shows a successful connection to google.com

You can also use the IP address to ping directly.

Functions:

The command is used to measure the average response. If there is no response for the ping command, you can assume one of the following issues with the network:

- There is a physical issue causing network loss.
- The destination address might be dysfunctional or incorrect.
- The ping request is blocked due to a target.
- There might be a problem with the routing table.

5. netstat

Linux netstat command refers to the network statistics. It provides statistical figures about different interfaces which include open sockets, routing tables, and connection information.

Syntax:

```
netstat -v
```

Output:

```

root@DESKTOP-PNM9727:~/S2MCA# netstat -v
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type        State         I-Node    Path
netstat: no support for `AF IPX' on this system.
netstat: no support for `AF AX25' on this system.
netstat: no support for `AF X25' on this system.
netstat: no support for `AF NETROM' on this system.
netstat: no support for `AF ROSE' on this system.
root@DESKTOP-PNM9727:~/S2MCA#

```

6. ss

Linux ss command is the replacement for netstat command. It is regarded as a much faster and more informative command than netstat. This command gives information about all TCP, UDP, and UNIX socket connections. The faster response of ss is possible as it fetches all the information from within the kernel userspace.

Syntax: ss

Output:

```

root@DESKTOP-PW0172Z:~/S2HCA# ss
Netid      State      Recv-Q     Send-Q     Local Address:Port      Peer Address:Port      Process
???
```

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
icmp6	UNCONN	0	0	0.0.0.0:icmp	0.0.0.0:*	
	UNCONN	0	0	*:1pv6-icmp	*:*	
v_str	ESTAB	0	0	*:37316651	2:50000	
v_str	ESTAB	0	0	*:37316652	2:50000	
v_str	ESTAB	0	0	*:37316653	2:50001	
v_str	ESTAB	0	0	*:37316654	2:50000	
v_str	ESTAB	0	0	*:37316656	2:50000	
v_str	ESTAB	0	0	*:37316657	2:50001	
v_str	ESTAB	0	0	*:37316658	2:50001	
v_str	ESTAB	0	0	*:37316659	2:50002	
v_str	ESTAB	0	0	*:37316660	2:50002	
v_str	ESTAB	0	0	*:37316661	2:50002	
v_str	ESTAB	0	0	*:37316662	2:50002	
v_str	ESTAB	0	0	*:37316664	2:574576650	
v_str	ESTAB	0	0	*:37316665	2:574576651	
v_str	ESTAB	0	0	*:37316666	2:574576653	
v_str	ESTAB	0	0	*:37316665	2:574576652	
v_str	ESTAB	0	0	*:37316665	2:574576651	

```

root@DESKTOP-PW0172Z:~/S2HCA#
```

7. dig

Linux dig command stands for Domain Information Groper. This command is used in DNS lookup to query the DNS name server. It is also used to troubleshoot DNS related issues.

It is mainly used to verify DNS mappings, MX Records, host addresses, and all other DNS records for a better understanding of the DNS topography.

This command is an improvised version of nslookup command.

Syntax: dig <domainName>

Example: dig geeksforgeeks.org

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# dig geeksforgeeks.org

;<>> DiG 9.18.28-0ubuntu0.22.04.1-Ubuntu <>> geeksforgeeks.org
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 12459
;; flags: qr rd ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;geeksforgeeks.org.          IN      A

;; ANSWER SECTION:
geeksforgeeks.org.          0       IN      A      34.218.62.116

;; Query time: 90 msec
;; SERVER: 172.21.64.1#53(172.21.64.1) (UDP)
;; WHEN: Sat May 24 07:37:46 IST 2025
;; MSG SIZE rcvd: 68

root@DESKTOP-PNM9727:~/S2MCA#
```

8. nslookup

Linux nslookup is also a *command* used for DNS related queries. It is the older version of dig.

Syntax: nslookup <domainName>

Example: nslookup www.google.com

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# nslookup google.com
;; Got recursion not available from 172.21.64.1
Server:          172.21.64.1
Address:         172.21.64.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.195.14
;; Got recursion not available from 172.21.64.1
Name:   google.com
Address: 2404:6800:4002:817::200e

root@DESKTOP-PNM9727:~/S2MCA#
```

9. route

Linux route command displays and manipulates the routing table existing for your system. A router is basically used to find the best way to send the packets across to a destination.

Syntax: route

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# route
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
default        DESKTOP-PNM9727 0.0.0.0         UG    0      0      0 eth0
172.21.64.0    0.0.0.0        255.255.240.0   U      0      0      0 eth0
root@DESKTOP-PNM9727:~/S2MCA#
```

10. host

Linux host command displays the domain name for a given IP address and IP address for a given hostname. It is also used to fetch DNS lookup for DNS related query. You can combine the host command with -t, and get DNS resource records like SOA, NS, A, PTR, CNAME, MX, SRV.

Syntax: host <resourceName>

Example: host

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# host
Usage: host [-aCdilrTvVw] [-c class] [-N ndots] [-t type] [-W time]
          [-R number] [-m flag] [-p port] hostname [server]
  -a is equivalent to -v -t ANY
  -A is like -a but omits RRSIG, NSEC, NSEC3
  -c specifies query class for non-IN data
  -C compares SOA records on authoritative nameservers
  -d is equivalent to -v
  -l lists all hosts in a domain, using AXFR
  -m set memory debugging flag (trace|record|usage)
  -N changes the number of dots allowed before root lookup is done
  -p specifies the port on the server to query
  -r disables recursive processing
  -R specifies number of retries for UDP packets
  -s a SERVFAIL response should stop query
  -t specifies the query type
  -T enables TCP/IP mode
  -U enables UDP mode
  -v enables verbose output
  -V print version number and exit
  -w specifies to wait forever for a reply
  -W specifies how long to wait for a reply
  -4 use IPv4 query transport only
  -6 use IPv6 query transport only
root@DESKTOP-PNM9727:~/S2MCA#
```

11. arp

Linux arp command stands for Address Resolution Protocol. It is used to view and add content to the kernel's ARP table. All the systems maintain a table of IP addresses and their corresponding MAC addresses. This table is called the ARP Lookup table. When a destination is requested to connect through IP address, your router will check for the MAC address in this table. If it is cached, the table will not be used.

Syntax: arp

Example: \$ arp

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# arp
Address                  HWtype  HWaddress     Flags Mask    Iface
DESKTOP-PNM9727.mshome. ether    00:15:5d:7a:f3:76  C           eth0
```

12. iwconfig

Linux iwconfig is used to configure the wireless network interface. It is used to set and view the basic WI-FI details like SSID and encryption. To know more about this command, refer to the man page.

Syntax : iwconfig

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# iwconfig
lo          no wireless extensions.

bond0       no wireless extensions.

dummy0      no wireless extensions.

eth0        no wireless extensions.

tunl0       no wireless extensions.

sit0        no wireless extensions.

root@DESKTOP-PNM9727:~/S2MCA#
```

13. hostname

Linux hostname is the simple command used to view and set the hostname of a system. The hostname set through this command is not permanent. It will be reset to the name in the hostname file back when the system reboots.

Syntax:hostname

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# hostname
DESKTOP-PNM9727
```

14. curl & wget

Linux curl and wget commands are used in downloading files from the internet through CLI. The curl command has to be used with the option "O" to fetch the file, while the wget command is used directly. Below are the syntax and the example for the two commands.

a) Curl

Syntax: curl -O<fileLink>

Example: curl -o google.com/doodles/childrens-day-

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# curl -O google.com/doodles/childrens-day-
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 241 100 241  0  0 542  0 --:--:-- --:--:-- --:--:-- 545
```

b) wget

syntax: wget<fileLink>

Example: wget google.com/doodles/new-years-day 2012

Output:

```
root@DESKTOP-PNM9727:~/S2MCA# wget https://www.google.com/doodles/new-years-day-2012
--2025-05-24 19:34:09-- https://www.google.com/doodles/new-years-day-2012
Resolving www.google.com (www.google.com)... 142.251.221.196, 2404:6800:4007:838::2004
Connecting to www.google.com (www.google.com)[142.251.221.196]:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://doodles.google/doodle/new-years-day-2012/ [following]
--2025-05-24 19:34:09-- https://doodles.google/doodle/new-years-day-2012/
Resolving doodles.google (doodles.google)... 142.250.183.145, 2404:6800:4007:831::2011
Connecting to doodles.google (doodles.google)[142.250.183.145]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 95830 (94K) [text/html]
Saving to: 'new-years-day-2012'

new-years-day-2012 100%[=====] 93.58K
2025-05-24 19:34:10 (345 KB/s) - 'new-years-day-2012' saved [95830/95830]
```

15. mtr

Linux mtr command is a combination of ping and the traceroute command. It continuously displays information regarding the packets sent with the ping time of each hop. It is also used to view the network issues.

Syntax: mtr<path>

Output:

```
DESKTOP-PHM9727 (172.21.64.206) -> www.google.com (142.251.221.196) My traceroute [v0.95] 2025-05-24T07:44:27+0530
Keys: Help Display mode Restart statistics Order of fields quit

Host      Packets  Loss%  Snt  Last  Avg  Best  Wrst StDev
1. DESKTOP-PHM9727.mshome.net  0.0%    8    0.4    0.8    0.4    1.1    0.2
2. 192.168.1.1                0.0%    8    3.9    4.5    3.8    6.6    0.9
3. 10.250.0.0                 0.0%    8   11.6    8.0    6.3   11.6    1.9
4. (waiting for reply)
5. (waiting for reply)
6. (waiting for reply)
7. (waiting for reply)
8. (waiting for reply)
9. pnmaaa-ba-in-f4.1e100.net   0.0%    7   23.3  166.8   20.6  1029. 380.5
```

Result

Completely study about basic Network commands

EXPERIMENT 7

CONCEPT OF SUBNETS, CIDR ADDRESS SCHEMES, SUBNET MASKS, IPTABLES

IPv4 address:

Every host and router on internet has an IP address which encodes its network number and host number, combination of the two is unique. An IP address does not actually refer to a host but refers to a network interface, so if host is on two networks, it must have two IP addresses.

Default Mask:

An address mask determines which portion of an IP address represents network number and which part represents host number, e.g., IP address, the mask has four octets. If a given bit of the mask is 1, the corresponding bit of the IP address is in network portion and if given bit of mask is 0, the corresponding bit of IP address is in host portion.

CIDR:

Classless Inter Domain Routing uses slash(/) notation to specify the mask with IPv4 address. The address is given as x.y.z.t/n where x.y.z.t is the IP address and n is the number of 1's in the default mask.

Network address:

First address of a network is network address. It is obtained by ANDing the mask with the IP address (Both in binary form). Another method is to set last 32-n bits of the IP address to 0.

Broadcast address:

Last address of a network is broadcast address. It is obtained by ORing the complement mask with the IP address (Both in binary form). Another method is to set last 32-n bits of the IP address to 1.

Class:

Class of an address is identified by the first byte of address. There are currently five classes A, B, C, D and E. The range of first byte of each class is:

Class A: 0 - 127

Class B: 128 - 191

Class C: 192 - 223

Class D: 224 - 239

Class E: 240 – 255

Subnetting using ipcalc

When managing a network, you will undoubtedly need to deal with subnetting. Some network administrators are able to do the binary math quite quickly in their head, to determine the subnet mask. However, others may need some help and this is where the **ipcalc** tool comes in handy.

To install **ipcalc**, simply run one of the commands below, based on the Linux distribution you are using.

\$ sudo apt install ipcalc

Get information about the network address:

#ipcalc 192.168.20.0

Output

```
root@LAPTOP-1B2435NS:~# cd S2MCA
root@LAPTOP-1B2435NS:~/S2MCA# ipcalc 192.168.20.0
Address:    192.168.20.0      11000000.10101000.00010100. 00000000
Netmask:    255.255.255.0 = 24 11111111.11111111.11111111. 00000000
Wildcard:   0.0.0.255        00000000.00000000.00000000. 11111111
=>
Network:    192.168.20.0/24   11000000.10101000.00010100. 00000000
HostMin:    192.168.20.1      11000000.10101000.00010100. 00000001
HostMax:    192.168.20.254    11000000.10101000.00010100. 11111110
Broadcast:  192.168.20.255    11000000.10101000.00010100. 11111111
Hosts/Net:  254               Class C, Private Internet
```

Calculate a subnet for **192.168.20.0/24**.

ipcalc 192.168.20.0/24

Output

```
root@LAPTOP-1B2435NS:~/S2MCA# ipcalc 192.168.20.0/24
Address:    192.168.20.0      11000000.10101000.00010100. 00000000
Netmask:    255.255.255.0 = 24 11111111.11111111.11111111. 00000000
Wildcard:   0.0.0.255        00000000.00000000.00000000. 11111111
=>
Network:    192.168.20.0/24   11000000.10101000.00010100. 00000000
HostMin:    192.168.20.1      11000000.10101000.00010100. 00000001
HostMax:    192.168.20.254    11000000.10101000.00010100. 11111110
Broadcast:  192.168.20.255    11000000.10101000.00010100. 11111111
Hosts/Net:  254               Class C, Private Internet
```

Calculate a single subnet with 10 hosts:

```
# ipcalc 192.168.20.0 -s 10
```

Output

```
root@LAPTOP-1B2435NS:~/S2MCA# ipcalc 192.168.20.0 -s 10
Address: 192.168.20.0      11000000.10101000.00010100. 00000000
Netmask: 255.255.255.0 = 24 11111111.11111111.11111111. 00000000
Wildcard: 0.0.0.255      00000000.00000000.00000000. 11111111
=>
Network: 192.168.20.0/24   11000000.10101000.00010100. 00000000
HostMin: 192.168.20.1     11000000.10101000.00010100. 00000001
HostMax: 192.168.20.254   11000000.10101000.00010100. 11111110
Broadcast: 192.168.20.255 11000000.10101000.00010100. 11111111
Hosts/Net: 254           Class C, Private Internet

1. Requested size: 10 hosts
Netmask: 255.255.255.240 = 28 11111111.11111111.11111111.1111 0000
Network: 192.168.20.0/28     11000000.10101000.00010100.0000 0000
HostMin: 192.168.20.1       11000000.10101000.00010100.0000 0001
HostMax: 192.168.20.14      11000000.10101000.00010100.0000 1110
Broadcast: 192.168.20.15    11000000.10101000.00010100.0000 1111
Hosts/Net: 14               Class C, Private Internet

Needed size: 16 addresses.
Used network: 192.168.20.0/28
Unused:
192.168.20.16/28
192.168.20.32/27
192.168.20.64/26
192.168.20.128/25
```

If you want to suppress the binary output, you can use the -b option as shown.

```
# ipcalc -b 192.168.20.100
```

Output

```
root@LAPTOP-1B2435NS:~/S2MCA# ipcalc -b 192.168.20.100
Address: 192.168.20.100
Netmask: 255.255.255.0 = 24
Wildcard: 0.0.0.255
=>
Network: 192.168.20.0/24
HostMin: 192.168.20.1
HostMax: 192.168.20.254
Broadcast: 192.168.20.255
Hosts/Net: 254           Class C, Private Internet
```

Result

Study about subnetting using ipcalc too

EXPERIMENT 8

SETTING UP A FIREWALL FOR LAN

Firewalls are a vital part of network security. A firewall is able to manage this traffic by monitoring network ports. In the world of firewalls, the term *port* doesn't refer to a physical connection like a USB, VGA, or HDMI port. For the purpose of firewalls, a *port* is an artificial construct created by the operating system to represent a pathway for a specific type of data.

Uncomplicated Firewall, abbreviated as UFW, is an easy-to-use and readable firewall interface. It supports IPV4 and IPV6 versions of ubuntu.

Step 1: Update APT

First update your API. So, by entering the following commands:

\$ sudo apt update

Step 2: Upgrade APT

\$ sudo apt upgrade

Step 3: Download and Install UFW

\$ sudo apt install UFW

Step 4: Enable / Disable UFW

You can enable / disable the UFW service every time the system boots up via the following commands:

\$ sudo UFW enable

\$ sudo UFW disable

Checks the current status of UFW using the terminal command below:

\$ sudo UFW status

Step 5: Blocks incoming and allow outgoing data traffic to block any incoming and allow outgoing data traffic,

which is the default configuration in the UFW policy, enter the following commands:

```
$ sudo UFW default allow outgoing
```

Step 6: Add, modify and delete Firewall rules.

```
$ sudo UFW allow http
```

```
$ sudo UFW allow 80
```

You can specify ranges for a specific port. To do so, execute the following commands:

```
$ sudo UFW allow 2000:3000/tcp
```

Result

Study above setting up a firewall for Lan.

EXPERIMENT 9

ANALYZING NETWORK PACKET USING WIRESHARK

Wireshark is one of the best network protocols for analyzing freely available packages. Previously known as Ethereal, Wireshark is widely used by industries and educational institutes.

Features

Wireshark contains several useful features, the foremost of which are listed below:

- Inspecting thousands of protocols
- New protocols being added with every update
- Live-capturing of protocols with offline analysis
- Three-way handshake

Step 1: Update APT

First, as always, update and upgrade your APT through the following command.

\$ sudo apt upgrade

Step 2: Download and Install Wireshark

\$ sudo apt install wireshark

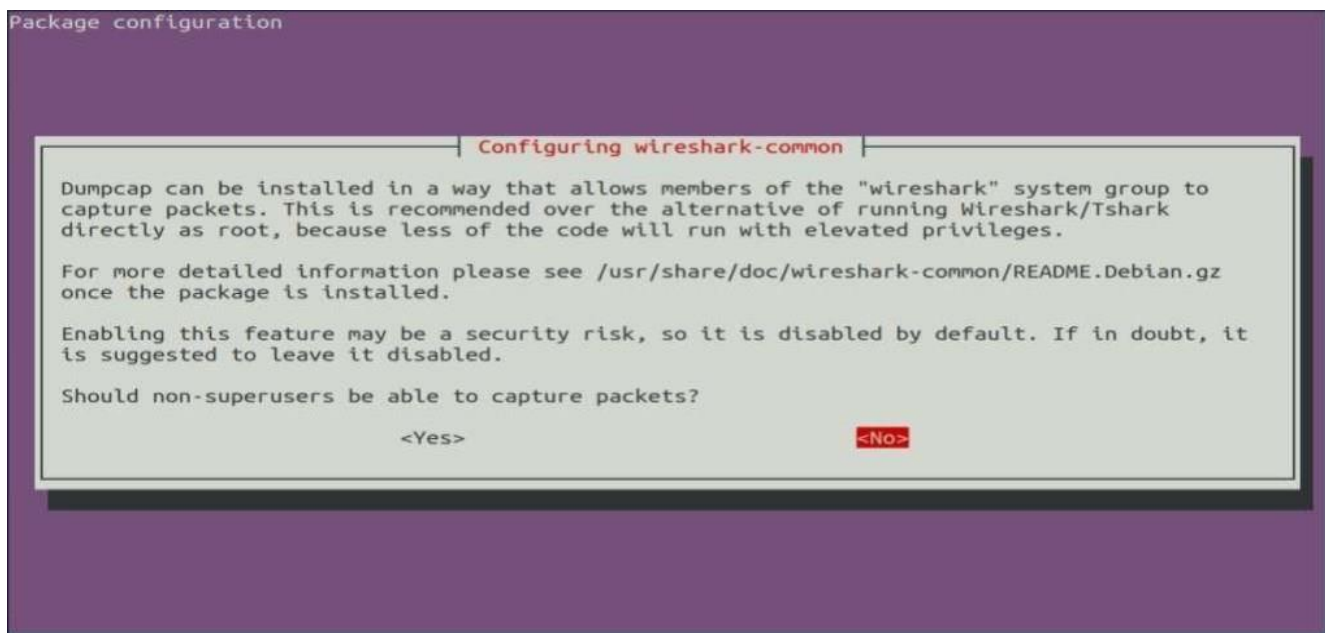
```

younits@younits-VirtualBox:~$ sudo apt install wireshark
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libc-ares2 libdouble-conversion3 liblua5.2-0 libpcr2-16-0 libqt5core5a libqt5dbus5 libqt5gui5
  libqt5multimedia5 libqt5multimedia5-plugins libqt5multimediagsttools5 libqt5multimediawidgets5
  libqt5network5 libqt5opengl5 libqt5sprintsupport5 libqt5svg5 libqt5widgets5 libsmi2ldbl
  libspandsp2 libwireshark-data libwireshark13 libwiretap10 libwsutil11 libxcb-xinerama0
  libxcb-xinput0 qt5-gtk-platformtheme qttranslations5-l10n wireshark-common wireshark-qt
Suggested packages:
  qt5-image-formats-plugins qtwayland5 snmp-mibs-downloader geoipupdate geoip-database
  geoip-database-extra libjs-leaflet libjs-leaflet.markercluster wireshark-doc
The following NEW packages will be installed:
  libc-ares2 libdouble-conversion3 liblua5.2-0 libpcr2-16-0 libqt5core5a libqt5dbus5 libqt5gui5
  libqt5multimedia5 libqt5multimedia5-plugins libqt5multimediagsttools5 libqt5multimediawidgets5
  libqt5network5 libqt5opengl5 libqt5sprintsupport5 libqt5svg5 libqt5widgets5 libsmi2ldbl
  libspandsp2 libwireshark-data libwireshark13 libwiretap10 libwsutil11 libxcb-xinerama0
  libxcb-xinput0 qt5-gtk-platformtheme qttranslations5-l10n wireshark wireshark-common wireshark-qt
0 upgraded, 29 newly installed, 0 to remove and 0 not upgraded.
Need to get 32.7 MB of archives.
After this operation, 163 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu focal/universe amd64 libdouble-conversion3 amd64 3.1.5-4ubuntu
1 [37.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu focal/main amd64 libpcr2-16-0 amd64 10.34-7 [181 kB]
Get:3 http://archive.ubuntu.com/ubuntu focal/universe amd64 libqt5core5a amd64 5.12.8+dfsg-0ubuntu1 [
2,005 kB]
5% [3 libqt5core5a 1,309 kB/2,005 kB 65%]

```

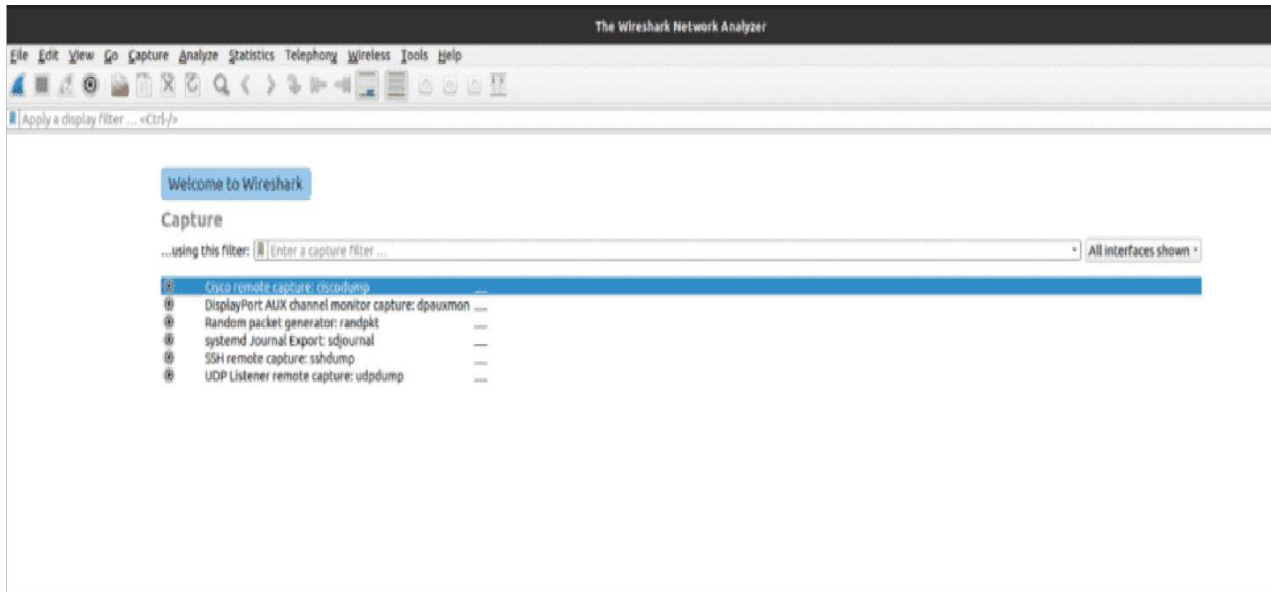
Step 3: Enable Root Privileges

When Wireshark installs on your system, you will be prompted by the following window. As Wireshark requires super user/root privileges to operate, this option asks to enable or disable permissions for all every user on the system. Press the “Yes” button to allow other users, or press the “No” button to restrict other users from using Wireshark.

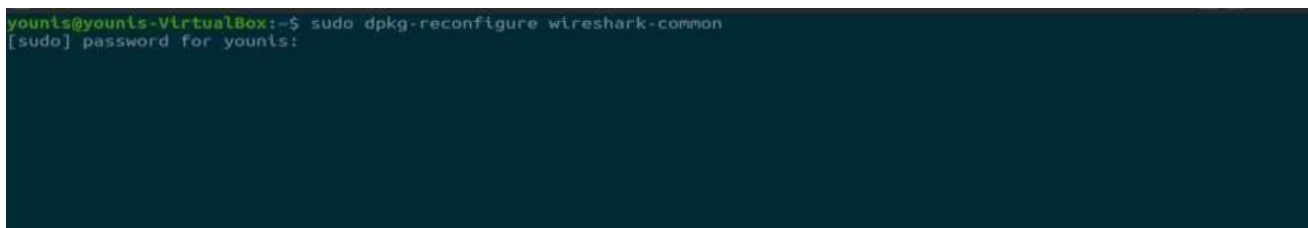


Step 4: (Optional) Reconfigure Permission Settings

If you have selected “No” in the above scenario, then you can change this selection again by executing the following command, which will reconfigure the Wireshark permission settings.



\$ sudo dpkg - reconfigure wireshark - common



Select the “Yes” button to change the configuration settings to allow other users access to Wireshark. In the terminal window, type the following command to start the Wireshark application.

\$ wireshark

Result

Study about analyzing network packet stream using Wireshark

EXPERIMENT 10

ANALYZING NETWORK PACKET STREAM USING TCPDUMP

TCPdump is a free & open-source packet analyzer tool & command-line utility. It is used for capturing the packets & inspect the network traffic going to & from our system. It is basic used for troubleshooting network issues & security testing. We can capture non-TCP traffic such as UDP, ARP or ICMP.

Step 1: Update the System.

sudo apt-get update

```
root@LAPTOP-1B2435NS:~/S2MCA# sudo apt-get update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.2 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:7 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [161 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [376 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7072 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [16.4 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 1015 kB in 2s (429 kB/s)
Reading package lists... Done
root@LAPTOP-1B2435NS:~/S2MCA#
```

Step 2: Install TCPdump on system.

sudo apt-get install tcpdump

```
root@LAPTOP-1B2435NS:~/S2MCA# sudo apt-get install tcpdump
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tcpdump is already the newest version (4.99.4-3ubuntu4).
0 upgraded, 0 newly installed, 0 to remove and 45 not upgraded.
root@LAPTOP-1B2435NS:~/S2MCA#
```

Check the TCPdump version.

sudo tcpdump --version

```
root@LAPTOP-1B2435NS:~/S2MCA# sudo tcpdump --version
tcpdump version 4.99.4
libpcap version 1.10.4 (with TPACKET_V3)
OpenSSL 3.0.13 30 Jan 2024
root@LAPTOP-1B2435NS:~/S2MCA#
```

Step 3: TCPdump syntax & Examples.

Sudo tcpdump [options] [expression]

- To capture all packets.

tcpdump

Here is the command output.

- To capture packets from a specific interface.

tcpdump -i eth0

```
root@LAPTOP-1B2435NS:~/S2MCA# sudo tcpdump -i eth0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:19:02.674713 IP LAPTOP-1B2435NS.mshome.net.51105 > 239.255.255.250.1900: UDP, length 170
10:19:03.689607 IP LAPTOP-1B2435NS.mshome.net.51105 > 239.255.255.250.1900: UDP, length 170
10:19:04.699408 IP LAPTOP-1B2435NS.mshome.net.51105 > 239.255.255.250.1900: UDP, length 170
10:19:07.016151 IP LAPTOP-1B2435NS.mshome.net.mdns > mdns.mcast.net.mdns: 0 PTR (QM)? _spotify-social-listening._tcp.local. (54)
```

- To capture only 10 packets.

tcpdump -c 10

```
root@LAPTOP-1B2435NS:~/S2MCA# sudo tcpdump -c 10
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:41:55.563448 IP LAPTOP-1B2435NS.mshome.net.mdns > mdns.mcast.net.mdns: 0 PTR (QM)? _spotify-connect._tcp.local. (45)
10:41:55.564778 IP6 fe80::1f0e:b8e3:3be0:7d9a.mdns > ff02::fb.mdns: 0 PTR (QM)? _spotify-connect._tcp.local. (45)
10:41:56.061451 IP LAPTOP-1B2435NS.mshome.net.60487 > 239.255.255.250.1900: UDP, length 125
10:41:58.279415 IP 172.23.152.222.35755 > prod-ntp-3.ntpl.ps5.canonical.com.ntp: NTPv4, Client, length 48
10:41:58.508325 IP prod-ntp-3.ntpl.ps5.canonical.com.ntp > 172.23.152.222.35755: NTPv4, Server, length 48
10:42:02.525626 IP LAPTOP-1B2435NS.mshome.net.57621 > 172.23.159.255.57621: UDP, length 44
10:42:04.514084 ARP, Request who-has LAPTOP-1B2435NS.mshome.net tell 172.23.152.222, length 28
10:42:04.515019 ARP, Reply LAPTOP-1B2435NS.mshome.net is-at 00:15:5d:95:5b:2f (oui Unknown), length 28
10:42:28.034746 IP 0.0.0.0 > 255.255.255.255: hip 68
10:42:31.331678 IP LAPTOP-1B2435NS.mshome.net.57621 > 172.23.159.255.57621: UDP, length 44
10 packets captured
10 packets received by filter
0 packets dropped by kernel
```

- To list all the available interface.

tcpdump -D

```

root@LAPTOP-1B2435NS:~/S2MCA# sudo tcpdump -D
1.eth0 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
root@LAPTOP-1B2435NS:~/S2MCA#

```

- To capture packets from any interface.

```
tcpdump -i any
```

- To use -n option to disable the translation.

```
tcpdump -n
```

```

root@LAPTOP-1B2435NS:~# sudo tcpdump -n
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
03:49:38.227207 IP 172.23.152.222.33930 > 185.125.190.58.123: NTPv4, Client, length 48
03:49:38.485356 IP 185.125.190.58.123 > 172.23.152.222.33930: NTPv4, Server, length 48
03:49:43.933559 ARP, Request who-has 172.23.152.222 (00:15:5d:eb:b0:60) tell 172.23.144.1, length 28
03:49:43.933586 ARP, Reply 172.23.152.222 is-at 00:15:5d:eb:b0:60, length 28
03:49:43.979429 ARP, Request who-has 172.23.144.1 tell 172.23.152.222, length 28
03:49:43.979660 ARP, Reply 172.23.144.1 is-at 00:15:5d:95:5b:2f, length 28
03:49:56.632836 IP 0.0.0.0 > 255.255.255.255: ip-proto-139 68
03:50:01.637487 IP 172.23.144.1.57621 > 172.23.159.255.57621: UDP, length 44
03:50:11.436250 IP 172.23.152.222.47095 > 185.125.190.58.123: NTPv4, Client, length 48
03:50:11.660248 IP 185.125.190.58.123 > 172.23.152.222.47095: NTPv4, Server, length 48
03:50:14.632820 IP6 fe80::215:5dff:feeb:b060 > ff02::2: ICMP6, router solicitation, length 16
03:50:17.192813 ARP, Request who-has 172.23.144.1 tell 172.23.152.222, length 28
03:50:17.193074 ARP, Reply 172.23.144.1 is-at 00:15:5d:95:5b:2f, length 28
^C
13 packets captured
13 packets received by filter
0 packets dropped by kernel

```

- To store capture network interface packets into a file.

```
tcpdump -n -i any > file.out
```

```

root@LAPTOP-1B2435NS:~# sudo tcpdump -n -i any>file.out
tcpdump: data link type LINUX_SLL2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
^C20 packets captured
20 packets received by filter
0 packets dropped by kernel

```

- To display the capture packets while saving to a file .

```
tcpdump -n -l | tee file.out
```

- **To capture packet from a specific port number.**

`tcpdump -n port 22`

- **To capture packets from a source ip.**

`tcpdump -n src host ip-address`

- **To capture the traffic coming from any source to port 80.**

`tcpdump -n dst port 80`

- **To capture all HTTP traffic coming from a source IP address**

`sudo tcpdump -n -i any src 192.168.1.10 and tcp port 80`

Result

Study about analysing network packet stream using tcpdump

EXPERIMENT 11

TCPDUMP COMMANDS

1. sudo tcpdump -D: To see the list of network interfaces valuable on the system on which tcpdump can capture packets

Output:

```
root@LAPTOP-1B2435NS:~/S2MCA# sudo tcpdump -D
1.eth0 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
root@LAPTOP-1B2435NS:~/S2MCA#
```

2. sudo tcpdump -i eth0: To capture the packets following through a specific interface

Output:

```
root@LAPTOP-1B2435NS:~/S2MCA# sudo tcpdump -i eth0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:19:02.674713 IP LAPTOP-1B2435NS.mshome.net.51105 > 239.255.255.250.1900: UDP, length 170
10:19:03.689607 IP LAPTOP-1B2435NS.mshome.net.51105 > 239.255.255.250.1900: UDP, length 170
10:19:04.699408 IP LAPTOP-1B2435NS.mshome.net.51105 > 239.255.255.250.1900: UDP, length 170
10:19:07.016151 IP LAPTOP-1B2435NS.mshome.net.mdns > mdns.mcast.net.mdns: 0 PTR (QU)? _spotify-social-listening._tcp.local. (54)
```

3. sudo tcpdump -c 4 wlp1s0 -i: To show the details of last 4 packets

4. sudo tcpdump -XX -i eth0: To see the information of the ip address in terms of ipv6 addressing (in hexadecimal format)

Output:

```
root@LAPTOP-1B2435NS:~/S2MCA# sudo tcpdump -xx -i eth0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:23:01.163994 IP LAPTOP-1B2435NS.mshome.net.57621 > 172.23.159.255.57621: UDP, length 44
    0x0000: ffff ffff ffff 0015 5d95 5b2f 0800 4500
    0x0010: 0048 3c99 0000 8011 75dc ac17 9001 ac17
    0x0020: 9fff e115 e115 0034 e79a 5370 6f74 5564
    0x0030: 7030 30ff fadc 14ff 23f2 0001 0004 4895
    0x0040: c203 19dc 6049 15f7 9de1 0f6b 7dc3 9943
    0x0050: 4b8c 2483 112b
10:23:01.402667 IP LAPTOP-1B2435NS.mshome.net.63668 > 239.255.255.250.1900: UDP, length 170
    0x0000: 0100 5e7f fffa 0015 5d95 5b2f 0800 4500
    0x0010: 00c6 ce97 0000 0111 be7c ac17 9001 efff
    0x0020: fffa f8b4 076c 00b2 7553 4d2d 5345 4152
    0x0030: 4348 202a 2048 5454 502f 312e 310d 0a48
    0x0040: 4f53 543a 2032 3339 2e32 3535 2e32 3535
    0x0050: 2e32 3530 3a31 3930 300d 0a4d 414e 3a20
    0x0060: 2273 7364 703a 6469 7363 6f76 6572 220d
    0x0070: 0a4d 583a 2031 0d0a 5354 3a20 7572 6e3a
    0x0080: 6469 616c 2d6d 756c 7469 7363 7265 656e
    0x0090: 2d6f 7267 3a73 6572 7669 6365 3a64 6961
    0x00a0: 6c3a 310d 0a55 5345 522d 4147 454e 543a
    0x00b0: 2043 6872 6f6d 6975 6d2f 3133 342e 302e
    0x00c0: 3639 3938 2e31 3738 2057 696e 646f 7773
    0x00d0: 0d0a 0d0a
10:23:02.393258 IP LAPTOP-1B2435NS.mshome.net.63668 > 239.255.255.250.1900: UDP, length 170
    0x0000: 0100 5e7f fffa 0015 5d95 5b2f 0800 4500
    0x0010: 00c6 ce98 0000 0111 be7b ac17 9001 efff
    0x0020: fffa f8b4 076c 00b2 7553 4d2d 5345 4152
```


5. **sudo tcpdump -i wlp1s0 -c5 port 80:** To access information of packet from a specific port number

6. **sudo tcpdump -i wlp1s0 -c 10 w icmp.pcap :**To store packet information limited by lines, words

Output:

```
root@LAPTOP-1B2435NS:~/S2MCA# sudo tcpdump -i eth0 -c 10 -w icmp.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10 packets captured
17 packets received by filter
0 packets dropped by kernel
root@LAPTOP-1B2435NS:~/S2MCA# |
```

7.**sudo tcpdump -r icmp.pcap :**To read the content of a packet capture by using tcpdump

Output:

```
root@LAPTOP-1B2435NS:~/S2MCA# sudo tcpdump -r icmp.pcap
reading from file icmp.pcap, link-type EN10MB (Ethernet), snapshot length 262144
10:27:02.194914 IP LAPTOP-1B2435NS.mshome.net.57621 > 172.23.159.255.57621: UDP, length 44
10:27:02.393537 IP LAPTOP-1B2435NS.mshome.net.60602 > 239.255.255.250.1900: UDP, length 170
10:27:03.385573 IP LAPTOP-1B2435NS.mshome.net.60602 > 239.255.255.250.1900: UDP, length 170
10:27:04.370337 IP LAPTOP-1B2435NS.mshome.net.60602 > 239.255.255.250.1900: UDP, length 170
10:27:05.352459 IP LAPTOP-1B2435NS.mshome.net.60602 > 239.255.255.250.1900: UDP, length 170
10:27:14.420297 IP LAPTOP-1B2435NS.mshome.net.mdns > mdns.mcast.net.mdns: 0*- [0q] 1/0/2 PTR LAPTOP-1B2435NS._dosvc._tcp.local. (192)
```

EXPERIMENT 12

DOCKERS AND CONTAINERS

CONTAINERS

The idea of containers as a way of running applications in isolation has been around for more than a decade. In practical terms, containers provide a layer of abstraction, allowing applications to be packaged with all their dependencies so they can run in any environment.

By taking advantage of OS virtualization primitives such as namespaces and control groups (cgroups), containers completely isolate application processes and protect them from interference and leaks.

In addition, a containerized environment offers control over the allocation of hardware resources (CPU, memory, and disk) available to a container.

Containers are faster than virtual machines because they don't require a guest operating system in every instance. Instead, they simply leverage the features and resources of the host OS to provide a working foundation for each container.

If you want to learn more about containers, you can read [Containerization vs Virtualization: Making the Right Choice for Your Infrastructure](#).

BENEFITS OF CONTAINERS

Compared to other virtualization technologies, containers offer significant operational benefits, such as:

Lightweight

By sharing the machine OS kernel, containers eliminate the need for full OS instances for the applications. This reduces the size of container files and makes them more resource-efficient.

Portability and Platform Independence

Since all files, dependencies, and assets required by the application are included in the container, you can set up and initiate a container anywhere you need. This also means that you only need to write an app once and then run it on any compatible environment, including local devices, cloud, and on-premises environments, without re-configuring it.

DOCKER

Docker is a great platform for packaging an application or a set of services in containers. A Docker container contains all the libraries and components required to run the application. From a practical point of view, a container is like a simplified virtual machine that functions independently of the underlying operating system.

In Docker, containers are generated from images that contain a complete operating system and/or pre-installed applications.

You can find a Docker image for all popular applications, including OS, DBMS, CRM, and programming languages. We strongly recommend checking out the official Docker image repository for an image. In case you can't find it, you can create your own image and deploy it to your Docker platform.

Let's see how you can install Docker on Ubuntu 22.04 in a couple of easy steps.

Installing Docker on Ubuntu

Ubuntu is the number one platform for managing Docker or Kubernetes containers. This is because Ubuntu runs the containers at scale, it is fast, secure, and open-source, powering millions of machines worldwide.

This article will cover two options for installing Docker on Ubuntu:

- **From the official Docker repository** - ensuring the latest Docker version.
- **From the default Ubuntu repositories** - providing a simpler installation.

Follow the steps below and install Docker on Ubuntu 20.04 or Ubuntu 22.04 using the method of your choice.

Important: Make sure to remove any older Docker installations before installing a new one. Removing previous Docker versions doesn't delete the images, containers, volumes, or networks you have created. Run the following command to uninstall previous versions:

```
sudo apt-get remove docker docker-engine docker.io containerd runc
```

Installing Docker from the Official Repository (Option 1)

Install Docker from the official Docker repository to ensure you get the latest stable program version. To access the official Docker repository, add the new package source to Ubuntu and then install Docker. Follow the steps below:

Step 1: Update the Package Repository

Run the following command to update the system's package repository and ensure the latest prerequisite packages are installed:

```
sudo apt update
```

When prompted, enter your root password and press **Enter** to proceed with the update.

Step 2: Install Prerequisite Packages

The apt package manager requires a few prerequisite packages on the system to use packages over HTTPS. Run the following command to allow Ubuntu to access the Docker repositories over HTTPS:

```
sudo apt install apt-transport-https ca-certificates curl software-properties-common -y
```

```
bosko@pnap:~$ sudo apt install apt-transport-https ca-certificates curl software-properties-common -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
ca-certificates is already the newest version (20211016~20.04.1).
ca-certificates set to manually installed.
software-properties-common is already the newest version (0.99.9.8).
software-properties-common set to manually installed.
The following additional packages will be installed:
  libcurl4
The following NEW packages will be installed:
  apt-transport-https curl libcurl4
0 upgraded, 3 newly installed, 0 to remove and 5 not upgraded.
Need to get 163 kB/398 kB of archives.
After this operation, 1,283 kB of additional disk space will be used.
```

The command above:

- Allows **apt** to transfer files and data over https.
- Allows the system to check security certificates.
- Installs curl, a data-transfer utility.
- Adds scripts for software management.

Step 3: Add GPG Key

A GPG key verifies the authenticity of a software package. Add the Docker repository GPG key to your system by running:

curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo apt-key add -

```
bosko@pnap:~$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo apt-key add -
OK
```

The output should state **OK**, verifying the authenticity.

Step 4: Add Docker Repository

Run the following command to add the Docker repository to **apt** sources:

sudo add-apt-repository "deb [arch=amd64] https://download.docker.com/linux/ubuntu \$(lsb_release -cs) stable"

```
bosko@pnap:~$ sudo add-apt-repository "deb [arch=amd64] https://download.docker.com/linux/ubuntu focal stable"
Get:1 https://download.docker.com/linux/ubuntu focal InRelease [57.7 kB]
Hit:2 http://us.archive.ubuntu.com/ubuntu focal InRelease
Get:3 http://security.ubuntu.com/ubuntu focal-security InRelease [114 kB]
Get:4 https://download.docker.com/linux/ubuntu focal/stable amd64 Packages [18.5 kB]
Get:5 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease [114 kB]
Get:6 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease [108 kB]
Get:7 http://security.ubuntu.com/ubuntu focal-security/main amd64 DEP-11 Metadata [40.7 kB]
Get:8 http://security.ubuntu.com/ubuntu focal-security/universe amd64 DEP-11 Metadata [77.6 kB]
Get:9 http://us.archive.ubuntu.com/ubuntu focal-updates/main amd64 DEP-11 Metadata [278 kB]
Get:10 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 DEP-11 Metadata [2,464 B]
```

The command adds the official Docker repository and updates the package database with the latest Docker packages.

Step 5: Specify Installation Source

Execute the **apt-cache** command to ensure the Docker installation source is the Docker repository, not the Ubuntu repository. The **apt-cache** command queries the package cache of the **apt** package manager for the Docker packages we have previously added.

Run the following command:

apt-cache policy docker-ce

```
bosko@pnap:~$ apt-cache policy docker-ce
docker-ce:
  Installed: (none)
  Candidate: 5:20.10.18~3-0~ubuntu-focal
  Version table:
   5:20.10.18~3-0~ubuntu-focal 500
        500 https://download.docker.com/linux/ubuntu focal/stable amd64 Packages
   5:20.10.17~3-0~ubuntu-focal 500
        500 https://download.docker.com/linux/ubuntu focal/stable amd64 Packages
   5:20.10.16~3-0~ubuntu-focal 500
        500 https://download.docker.com/linux/ubuntu focal/stable amd64 Packages
   5:20.10.15~3-0~ubuntu-focal 500
        500 https://download.docker.com/linux/ubuntu focal/stable amd64 Packages
   5:20.10.14~3-0~ubuntu-focal 500
```

Step 6: Install Docker

Install Docker by running:

sudo apt install docker-ce -y


```

bosko@pnap:~$ sudo apt install docker-ce -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  bridge-utils ubuntu-fan
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  containerd.io docker-ce-cli docker-ce-rootless-extras docker-scan-plugin
  slirp4netns
Suggested packages:
  aufs-tools cgroupfs-mount | cgroup-lite
The following packages will be REMOVED:
  containerd docker.io runc
The following NEW packages will be installed:
  containerd.io docker-ce docker-ce-cli docker-ce-rootless-extras
  docker-scan-plugin slirp4netns
0 upgraded, 6 newly installed, 3 to remove and 5 not upgraded.
Need to get 102 MB of archives.
After this operation, 64.1 MB of additional disk space will be used.

```

Wait for the installation process to complete.

Step 7: Check Docker Status

Check if Docker is installed, the daemon started, and the process is enabled to start on boot. Run the following command:

sudo systemctl status docker

```

bosko@pnap:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset:
   Active: active (running) since Thu 2022-10-06 18:17:02 EDT; 3min 52s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 2735 (dockerd)
      Tasks: 9
     Memory: 28.2M
    CGroup: /system.slice/docker.service
            └─2735 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/cont

```

The output states that the Docker daemon is up and running.